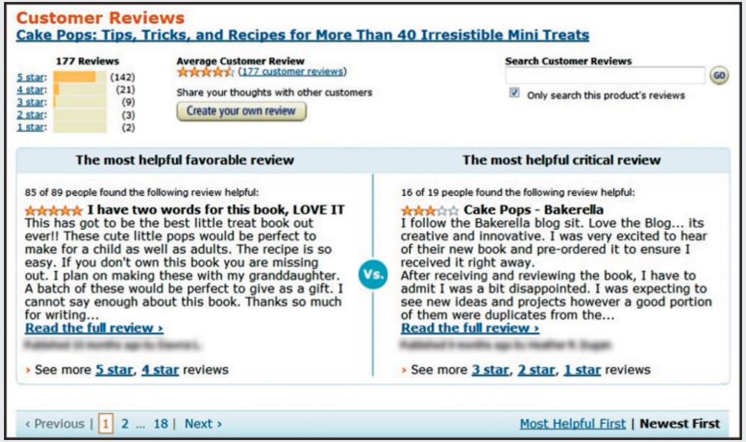
Chapter 2
Data Preparation

What is Data Preparation?

Unstructured Data

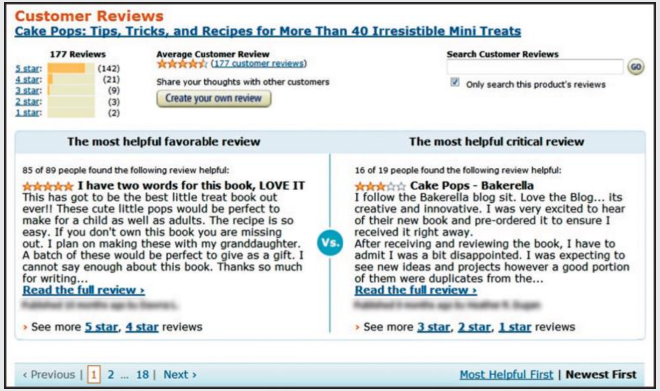


Structured Data

First Name	Age	GPA	Favorite Pet	Voted
Selma	18	3.7	Dog	0
Jerry	19	3.5	Cat	1
Sanjiv	17	3.1	Dog	1
Olivia	18	3.9	Other	0
Hector	18	3.6	Fish	0

What is Data Preparation?

- **Data preparation** is the process of collecting, transforming and storing a data set in a form that's useful for analysis
  - **Data collection**
  - **Data wrangling**
  - **Data management**



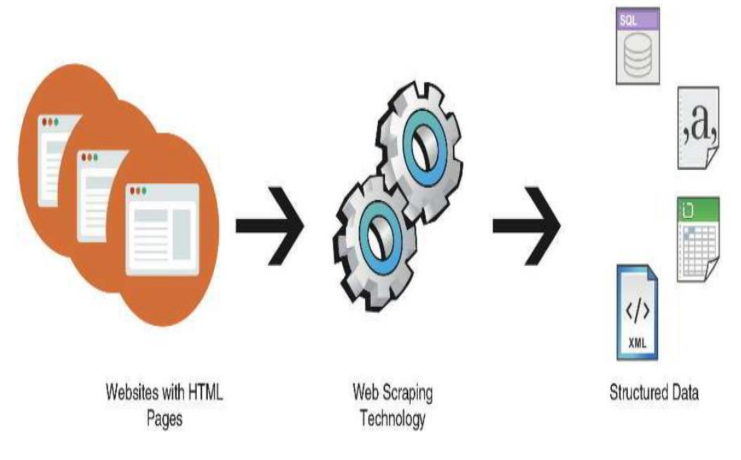
Data Preparation – 3 Steps

- **Data collection:** gathering relevant data from various sources
- **Data wrangling:** converting original data to a more usable format by cleaning, transforming, and integrating data to facilitate effective analysis and storytelling
- **Data management:** storing data to make it available for analysis

Data Collection

➡ Data **sources**:
- Websites
- Surveys
- Logs of scientific experiments or field research

Example: *Recording bird sounds and behavior in the wild*



➡ Data **collection methods**:
- Observational studies
- Experiments

Example: *Observing birds in the wild is an observational study*

Data Wrangling/Munging

Converting data into forms necessary for analysis and storytelling

1. Data Cleaning

2. Data Transformation

3. Data Integration

1. Data Cleaning

Fixing or removing incomplete, abnormal, or inconsistent data

➡ *Important:* **Report any adjustments** made to data during cleaning!

2. Data Transformation

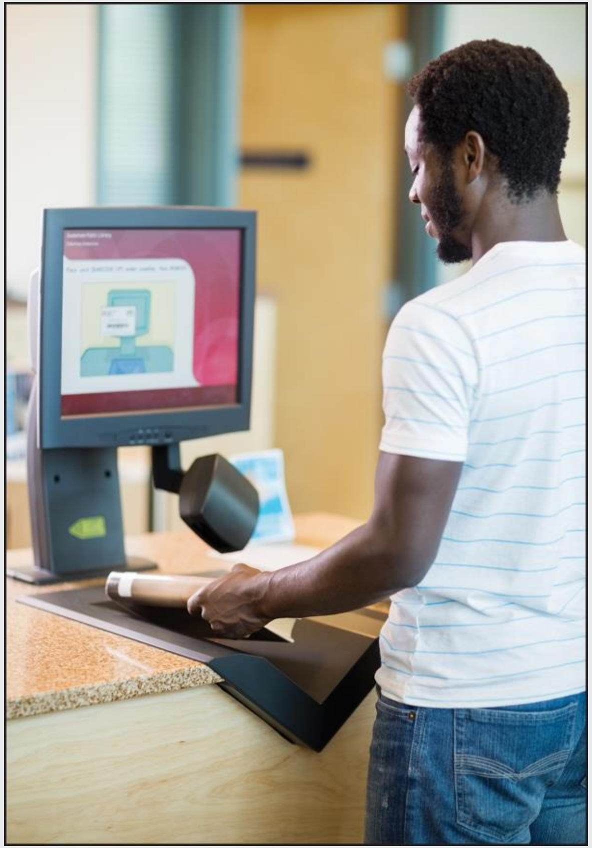
Organizing and formatting data for efficient analysis, including:

- Structuring previously unstructured data
- Further structuring a structured dataset

3. Data Integration

Combining two or more data sets to provide more insights

- When data are easily obtained from a public source, the competitive advantage of the information is low



Data Management

Storing data and facilitating its access

Summary

What is Data Preparation?

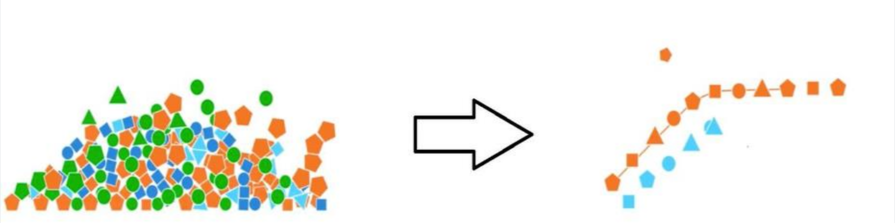
Data Collection

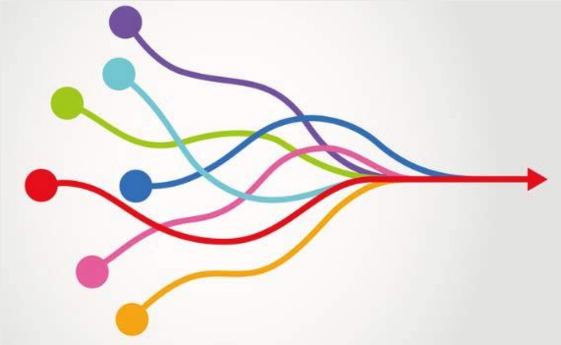
Data Wrangling:
- Cleaning
- Transforming
- Integrating

Data Management

Cleaning the data

Transforming the Data





Integrating the Data

Cleaning the data

- Missing data
- Implausible or extreme values
- Incorrectly formatted values
- Duplicate records

What is your favorite pet?

Your answer

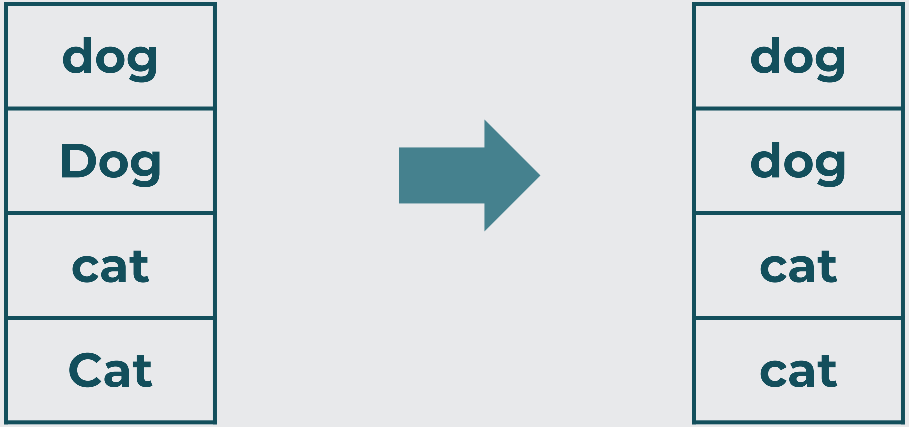
What is your favorite pet?

Your answer

dog
Dog
cat
Cat

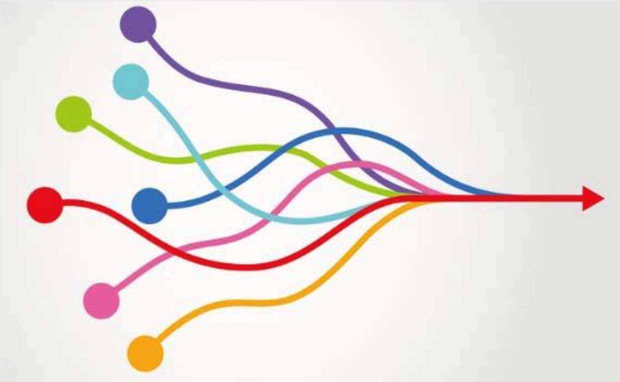
What is your favorite pet?

Your answer

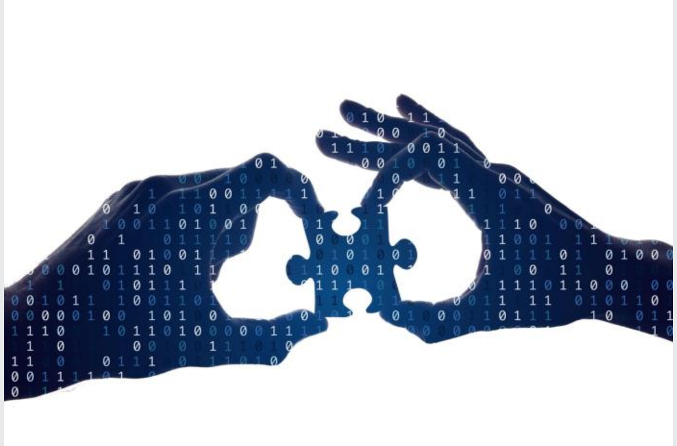


Integrating the data

- Stacking data sets
- Augmenting a data set
- Merging data sets



Overview



01 Cleaning Data

02 Missing Data

03 Imputation with Internal Data to Address Missing Data

04 Imputation with External Data and Other Methods to Address Missing Data

Overview

01 **Cleaning Data**

The Importance of Cleaning Data

- Cleaning data ensures data accuracy and reliability for analysis
- Prevents errors and biases that can lead to incorrect conclusions

Cleaning Data with Spreadsheet Functions

1

Utilize functions in software tools to streamline data cleaning processes

2

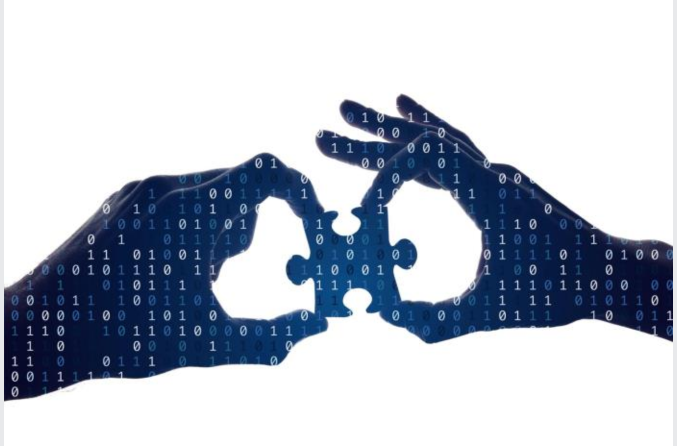
Sorting data allows for organizing information in a specific order for better analysis

3

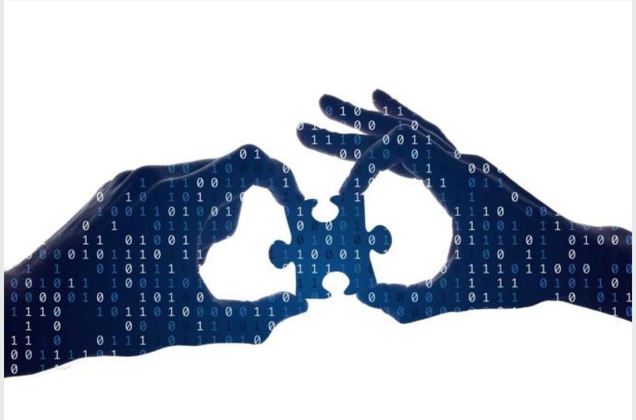
Filtering data helps in focusing on specific criteria or removing unwanted data entries

Overview

02 Missing Data



Missing Data



- Definition: values in a dataset that are absent but expected to be present
- Occurs because of data entry errors, equipment malfunctions, participant non-response, etc.
- Can lead to biased results and inaccurate statistical conclusions

Example of Missing Data

first_name	age	gpa	favorite_pet	voted
Selma	18	3.1	Dog	0
Juan		3.2	Cat	1
Rozzlin	19	3.9		0

Addressing Missing Data

Remove Observations

Delete rows with missing data

Impute with Internal Data

Fill in missing values based on data in the dataset

Impute with External Data

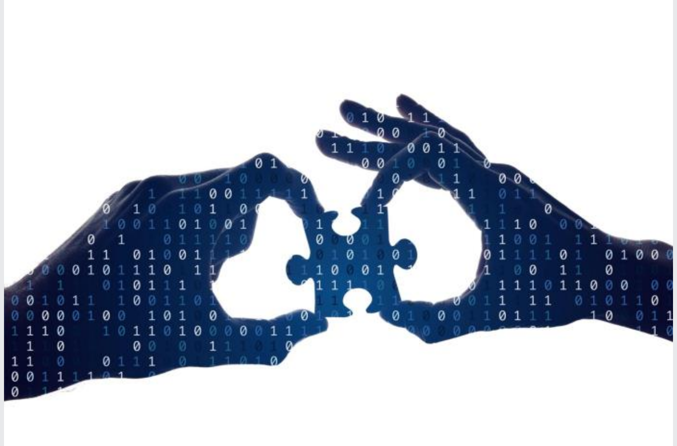
Utilize external sources or datasets to impute missing values

Other Options

Mean imputation, regression imputation, machine learning

Overview

03 Imputation with Internal Data to Address Missing Data



Imputation from Internal Data

- Replaces missing values with estimates derived from comparable data points
- Methods include using means, medians, or advanced techniques
- Utilizes data within the dataset to estimate missing values

object_id	type	material	depth	age
O1	Pottery	Clay	1	2000
O2	Jewelry	Gold	2	
O3	Tool	Bronze	1.5	3000
O4	Weapon	Iron	3.5	1500
O5	Coin	Silver	1.5	
O6	Pottery	Clay	2.5	2500
O7	Jewelry	Gold	1	
O8	Weapon	Bronze	3	2000
O9	Tool	Iron	4	
O10	Coin	Silver	1.2	1900
O11	Jewelry	Gold	2.7	
O12	Pottery	Clay	2.2	2400

Addressing Missing Data: Removing Observations or Internal Imputation

Method 1: Remove observations with missing values

object_id	type	material	depth	age
O1	Pottery	Clay	1	2000
O2	Jewelry	Gold	2	
O3	Tool	Bronze	1.5	3000
O4	Weapon	Iron	3.5	1500
O5	Coin	Silver	1.5	
O6	Pottery	Clay	2.5	2500
O7	Jewelry	Gold	1	
O8	Weapon	Bronze	3	2000
O9	Tool	Iron	4	
O10	Coin	Silver	1.2	1900
O11	Jewelry	Gold	2.7	
O12	Pottery	Clay	2.2	2400

Addressing Missing Data: Removing Observations or Internal Imputation

Method 1: Remove observations with missing values

(2000 + 3000 + 1500 + 2500 + 2000 + 1900 + 2400) / 7 = **2185.72**

object_id	type	material	depth	age
O1	Pottery	Clay	1	2000
O2	Jewelry	Gold	2	**2185.72**
O3	Tool	Bronze	1.5	3000
O4	Weapon	Iron	3.5	1500
O5	Coin	Silver	1.5	**2185.72**
O6	Pottery	Clay	2.5	2500
O7	Jewelry	Gold	1	**2185.72**
O8	Weapon	Bronze	3	2000
O9	Tool	Iron	4	**2185.72**
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O12	Pottery	Clay	2.2	2400

Addressing Missing Data: Removing Observations or Internal Imputation

Method 1: Remove observations with missing values

(2000 + 3000 + 1500 + 2500 + 2000 + 1900 + 2400) / 7 = **2185.72**

Method 2: Impute missing values

object_id	type	material	depth	age
O1	Pottery	Clay	1	2000
O2	Jewelry	Gold	2	**2185.72**
O3	Tool	Bronze	1.5	3000
O4	Weapon	Iron	3.5	1500
O5	Coin	Silver	1.5	**2185.72**
O6	Pottery	Clay	2.5	2500
O7	Jewelry	Gold	1	**2185.72**
O8	Weapon	Bronze	3	2000
O9	Tool	Iron	4	**2185.72**
O10	Coin	Silver	1.2	1900
O11	Jewelry	Gold	2.7	**2185.72**
O12	Pottery	Clay	2.2	2400

Addressing Missing Data: Removing Observations or Internal Imputation

Method 1: Remove observations with missing values

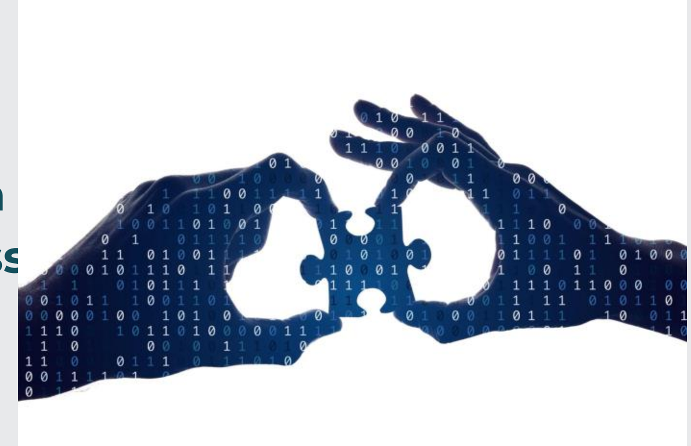
(2000 + 3000 + 1500 + 2500 + 2000 + 1900 + 2400) / 7 = **2185.72**

Method 2: Impute missing values

(2000 + 2185.72 + 3000 + 1500 + 2185.72 + 2500 + 2185.72 + 2000 + 2185.72 + 1900 + 2185.72 + 2400) / 12 = **2185.72**

Overview

04 Imputation with External Data
Other Methods to Address Miss



Imputation from External Data

- Uses values from a different dataset
- Substitute missing values for the same individual directly from another dataset
- Requires trust in the credibility and relevance of additional data sources
- Requires reporting decisions made during imputation

Other Options for Missing Data

Replace missing values with **unique identifiers**

Conduct **separate analyses** excluding missing data

Utilize **multiple imputation** techniques based on observed data patterns

Considerations

- Randomness in Data
- Documentation for Transparency
- Data Integrity
- Ethical Considerations

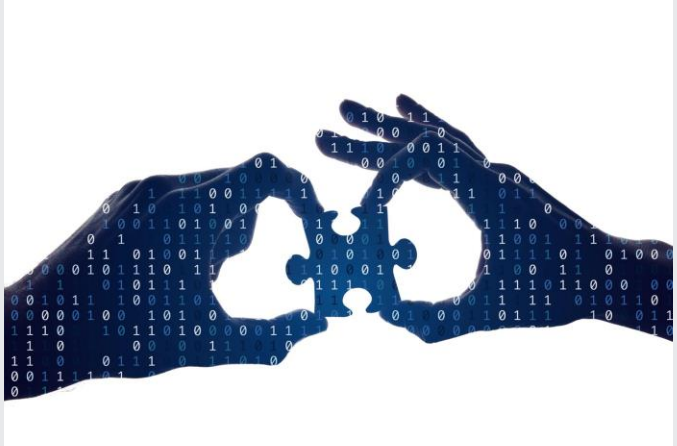
Summary

01 Cleaning Data

02 Missing Data

03 Imputation with Internal Data to Address Missing Data

04 Imputation with External Data and Other Methods to Address Missing Data



Overview

01 Anomalous Data Types

02 Implausible Values and Extreme Values

03 Incorrect Values and Duplicate Records

Overview

01 **Anomalous Data Types**

Types of Anomalous Values

- Implausible values
- Extreme values
- Incorrect data formats
- Duplicate records

Overview

02 Implausible Values and Extreme Values

Implausible Values

Credit Score Range	Credit Score Meaning
720 - 850	Excellent
690 - 719	Good
630 - 689	Fair
300 - 629	Bad

name	credit_score
Juan	800
Alice	2,995
Kai	690
Kahlil	-53

Pearson

Implausible Values

Credit Score Range	Credit Score Meaning
720 - 850	Excellent
690 - 719	Good
630 - 689	Fair
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name	credit_score
Juan	800
Alice	2,995
Kai	690
Kahlil	-53

Extreme Data Values and Outliers

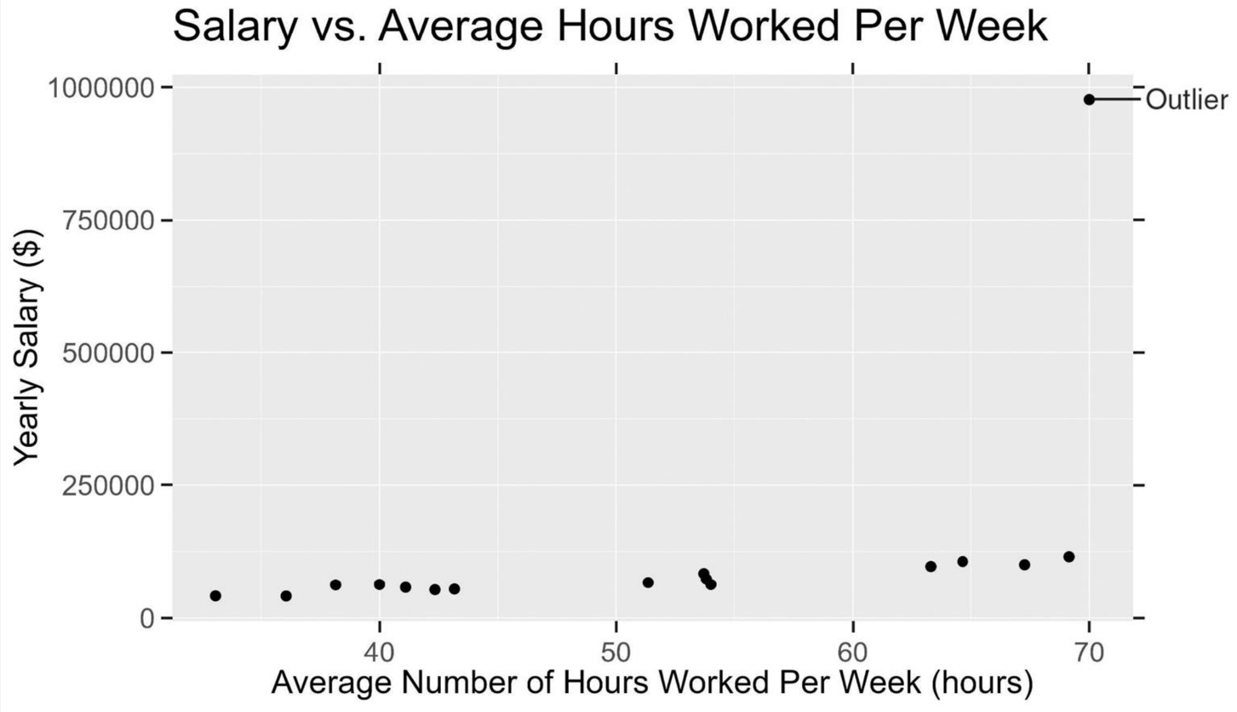
Extreme data values: Significantly deviate from the average or expected range

Response: Verify data accuracy, transform them, or filter based on domain knowledge

Outliers: May indicate errors or anomalies

Response: Use statistical techniques or remove outlier values from analysis

Extreme Value Likely to be an Outlier



Case Study: Dewey Defeats Truman

Case Study Overview:

- The 1948 presidential election famously had a newspaper declaring the wrong winner
- Identifying anomalous data could have identified and corrected the error before publication
- This case underscores the importance of thorough data cleaning processes in analytical work

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Overview

03 **Incorrect Values and Duplicate Records**

Incorrect Data Formats

Incorrect data formats:

- Numerical values stored as text
  “seventeen” instead of 17

- Inconsistent naming conventions
  “Goldfisg” instead of “Goldfish”

Deletion is a last resort

Identifying and Removing Duplicate Records

1

Identify potential duplicate records

2

Compare identification fields

3

Keep only one copy

Summary

01 Anomalous Data Types

02 Implausible Values and Extreme Values

03 Incorrect Values and Duplicate Records

Overview

Transforming Quantitative Variables

01 Creating a Categorical Variable from a Quantitative Variable

02 Changing the Scale of a Quantitative Variable

03 Combining Quantitative Variables

Overview

01 Creating a Categorical Variable from a Quantitative Variable

Quantitative and Categorical Variables

Quantitative Variables:

- Numerical
- Allow mathematical computations
- Discrete or continuous
- Examples: Age and GPA

Categorical Variables:

- Often non-numeric
- Represent group or category memberships
- Examples: Favorite Pet and Voted

First Name	Age	GPA	Favorite Pet	Voted
Selma	18	3.7	Dog	0
Jerry	19	3.5	Cat	1
Sanjiv	17	3.1	Dog	1
Olivia	18	3.9	Other	0
Hector	18	3.6	Fish	0

Creating a Categorical Variable from a Quantitative Variable

- Information in a quantitative variable may be more usefully represented as a category or label

- **For example**, for a restaurant that charges less for a child or an older adult, the quantitative age is less important than the label “child” or “senior”

First Name	Age	GPA	Favorite Pet	Voted
Selma	18	3.7	Dog	0
Jerry	19	3.5	Cat	1
Sanjiv	17	3.1	Dog	1
Olivia	18	3.9	Other	0
Hector	18	3.6	Fish	0

Creating a Categorical Variable from a Quantitative Variable

First Name	Age	GPA	Favorite Pet	Voted	Age Group
Selma	18	3.7	Dog	0	Adult
Jerry	19	3.5	Cat	1	Adult
Sanjiv	17	3.1	Dog	1	Adult
Olivia	18	3.9	Other	0	Adult
Hector	18	3.6	Fish	0	Adult

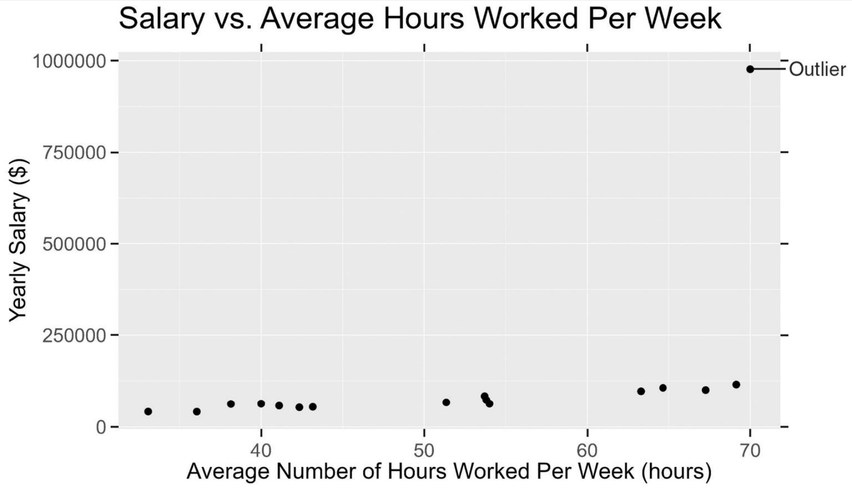
We categorize the quantitative variable Age in the following way:

- Ages 0-11: Child
- Ages 12-59: Adult
- Ages 60+: Senior

Overview

02 Changing the Scale of a Quantitative Variable

Multiply or Divide the Original Variable by Some Value



$20,000 / 1,000 = $20 (in thousands)
$800,000 / 1,000 = $800 (in thousands)

Dividing these salary values by 1,000 reduces the size of the values so they're easier to read!

Case Study: GlobalGiving

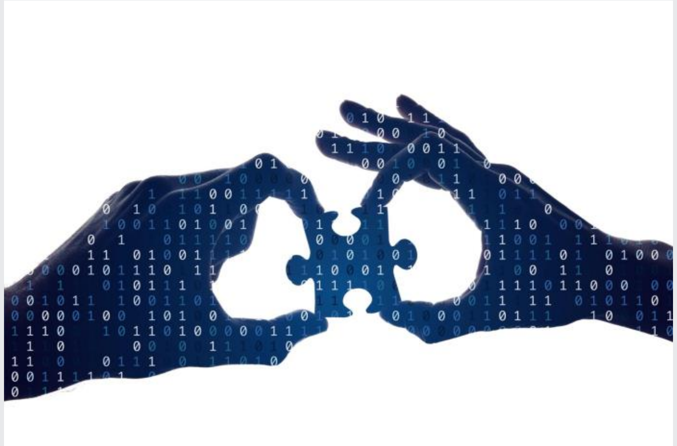
- **GlobalGiving** is a nonprofit that provides a platform for other nonprofits to raise funds
- **GlobalGiving** teaches nonprofits to centralize and clean their data, transforming quantitative and qualitative data

Overview

Combining Two or More Quantitative Variables

To find a country's GNI per capita, we use the equation:

GNI_ per_ capita = GNI / population



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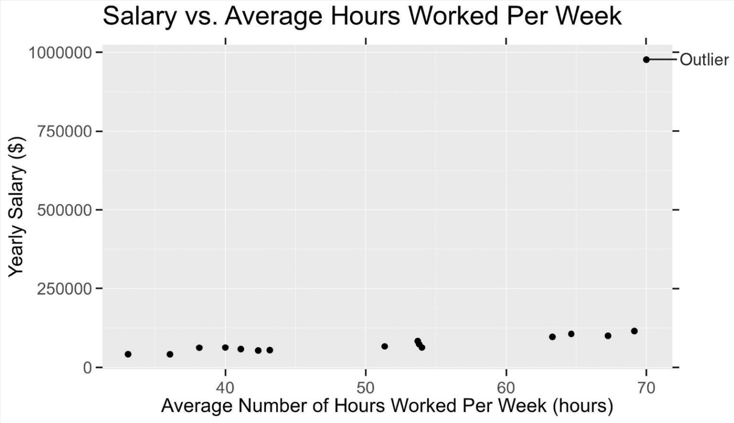
Summary



Creating a categorical variable from a quantitative Variable

Changing the scale of a quantitative variable

Combining quantitative variables

Overview

01 Creating Quantitative Variables and Encoding

02 Condensing Categories and Splitting

Overview

01 Creating Quantitative Variables and Encoding

Categorical Variables

- Represent different types of variables

  Examples: pet preference, date, class standing, political party, or zip code

- Often text variables (e.g. Favorite Pet), but may be numeric (e.g. Voted, or Zip Code)

- Not appropriate for mathematical operations

First Name	Age	GPA	Favorite Pet	Voted
Selma	18	3.7	Dog	0
Jerry	19	3.5	Cat	1
Sanjiv	17	3.1	Dog	1
Olivia	18	3.9	Other	0
Hector	18	3.6	Fish	0

Creating a Quantitative Variable from a Categorical Variable

- Many summarization, visualization, and analysis methods require numeric data

- Voted variable converted
  - “Yes” → 1
  - “No” → 0

First Name	Age	GPA	Favorite Pet	Voted
Selma	18	3.7	Dog	0
Jerry	19	3.5	Cat	1
Sanjiv	17	3.1	Dog	1
Olivia	18	3.9	Other	0
Hector	18	3.6	Fish	0

Creating a Quantitative Variable from a Categorical Variable

First Name	Age	GPA	Favorite Pet	Voted	Pet_Preference
Selma	18	3.7	Dog	0	1
Jerry	19	3.5	Cat	1	2
Sanjiv	17	3.1	Dog	1	1
Olivia	18	3.9	Other	0	4
Hector	18	3.6	Fish	0	3

- New **Pet_ Preference** variable - quantitative version of **Favorite Pet** variable
- **Pet_ Preference** associates each value of **Favorite Pet** with a numeric value

Encoding

- Variable **encoding**: Converting a categorical variable into a quantitative one by assigning numeric codes to values of the categorical variable
- **Boolean encoding**: Converting a categorical variable into a quantitative one by assigning a 1 or 0 to the values of the categorical variable

Response	Encoding Scheme 1	Encoding Scheme 2	Encoding Scheme 3	Encoding Scheme 4
Dislike	1	-1	0	0
Neutral	2	0	0	1
Like	3	1	1	1000

Overview

02 **Condensing Categories and Splitting**

Condense the Categories of a Categorical Variable

First Name	Age	GPA	Favorite Pet	Voted	Pet_Condense
Selma	18	3.7	Dog	0	Dog
Jerry	19	3.5	Cat	1	Cat
Sanjiv	17	3.1	Dog	1	Dog
Olivia	18	3.9	Other	0	Other
Hector	18	3.6	Fish	0	Other

- **Condensing categories**: Reducing the number of values in a categorical variable by merging categories into broader ones

- **Example:** Condense the categories of **Favorite Pet** down to “Dog”, “Cat”, and “Other”

Splitting a Variable into Multiple Variables

- Extracting information from the values of a categorical variable and storing the info in a new variable

- **Example:** Categorical variable Date contains piecies of information – *day, month, year*

Date	Month	Year
3/4/2003	March	2003
11/2/1998	November	1998
3/30/2013	March	2013
5/1/2009	May	2009
9/20/2012	September	2012

Summary

First Name	Age	GPA	Favorite Pet	Voted
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Olivia	18	3.9	Other	0
Hector	18	3.6	Fish	0

Creating a quantitative variable from a categorical variable

Condensing the categories of a categorical variable

Splitting a variable into multiple variables

Tidy Data

Reshaping a Data Set

Tidy data:

- Each column represents a distinct variable
- Each row represents a unique observation
- Each cell contains one value

DogID	Class	Name	Age	Breed
18654	Herding	Otis	10	BorderCollie
20862	Hound	Tito	5	Dachshund
12368	NonSporting	Chloe	15	Poodle
56731	NonSporting	Olive	1	Bulldog
90052	Sporting	Georgie	6	Labrador

Wide Format to Long Format

country	1800	1801	1802	1803	1804	1805	1806	1807	1808	1809	1810
Afghanistan	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M
Angola	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M
Albania	400k	402k	404k	405k	407k	409k	411k	413k	414k	416k	418k
Andorra	2650	2650	2650	2650	2650	2650	2650	2650	2650	2650	2650
UAE	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k
Argentina	534k	520k	506k	492k	479k	466k	453k	441k	429k	417k	420k
Armenia	413k	413k	413k	413k	413k	413k	413k	413k	413k	413k	413k
Antigua and Barbuda	37k	37k	37k	37k	37k	37k	37k	37k	37k	37k	37k
Australia	200k	205k	211k	216k	222k	227k	233k	239k	246k	252k	259k
Austria	3M	3.02M	3.04M	3.05M	3.07M	3.09M	3.11M	3.12M	3.14M	3.16M	3.18M
Azerbaijan	880k	880k	880k	880k	880k	880k	880k	880k	880k	880k	880k
Burundi	899k	899k	899k	899k	899k	899k	899k	899k	899k	899k	899k
Belgium	3.25M	3.26M	3.27M	3.28M	3.29M	3.3M	3.3M	3.31M	3.32M	3.33M	3.34M

country	year	population
Afghanistan	1800	3.28M
Afghanistan	1801	3.28M
Afghanistan	1802	3.28M
Afghanistan	1803	3.28M
Afghanistan	1804	3.28M
Afghanistan	1805	3.28M
Afghanistan	1806	3.28M
Afghanistan	1807	3.28M
Afghanistan	1808	3.28M
Afghanistan	1809	3.28M
Afghanistan	1810	3.28M
Afghanistan	1811	3.28M
Afghanistan	1812	3.28M
Afghanistan	1813	3.28M
Afghanistan	1814	3.28M
Afghanistan	1815	3.28M
Afghanistan	1816	3.28M
Afghanistan	1817	3.28M
Afghanistan	1818	3.28M
Afghanistan	1819	3.28M
Afghanistan	1820	3.29M
Afghanistan	1821	3.3M
Afghanistan	1822	3.31M
Afghanistan	1823	3.32M
Afghanistan	1824	3.34M

Long Format to Wide Format

country	year	population
Afghanistan	1800	3.28M
Afghanistan	1801	3.28M
Afghanistan	1802	3.28M
Afghanistan	1803	3.28M
Afghanistan	1804	3.28M
Afghanistan	1805	3.28M
Afghanistan	1806	3.28M
Afghanistan	1807	3.28M
Afghanistan	1808	3.28M
Afghanistan	1809	3.28M
Afghanistan	1810	3.28M
Afghanistan	1811	3.28M
Afghanistan	1812	3.28M
Afghanistan	1813	3.28M
Afghanistan	1814	3.28M
Afghanistan	1815	3.28M
Afghanistan	1816	3.28M
Afghanistan	1817	3.28M
Afghanistan	1818	3.28M
Afghanistan	1819	3.28M
Afghanistan	1820	3.29M
Afghanistan	1821	3.3M
Afghanistan	1822	3.31M
Afghanistan	1823	3.32M
Afghanistan	1824	3.34M

country	1800	1801	1802	1803	1804	1805	1806	1807	1808	1809	1810
Afghanistan	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M
Angola	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M
Albania	400k	402k	404k	405k	407k	409k	411k	413k	414k	416k	418k
Andorra	2650	2650	2650	2650	2650	2650	2650	2650	2650	2650	2650
UAE	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k
Argentina	534k	520k	506k	492k	479k	466k	453k	441k	429k	417k	420k
Armenia	413k	413k	413k	413k	413k	413k	413k	413k	413k	413k	413k
Antigua and Barbuda	37k	37k	37k	37k	37k	37k	37k	37k	37k	37k	37k
Australia	200k	205k	211k	216k	222k	227k	233k	239k	246k	252k	259k
Austria	3M	3.02M	3.04M	3.05M	3.07M	3.09M	3.11M	3.12M	3.14M	3.16M	3.18M
Azerbaijan	880k	880k	880k	880k	880k	880k	880k	880k	880k	880k	880k
Burundi	899k	899k	899k	899k	899k	899k	899k	899k	899k	899k	899k
Belgium	3.25M	3.26M	3.27M	3.28M	3.29M	3.3M	3.3M	3.31M	3.32M	3.33M	3.34M

Summary

country	1800	1801	1802	1803	1804	1805	1806	1807	1808	1809	1810
Afghanistan	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M
Angola	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M
Albania	400k	402k	404k	405k	407k	409k	411k	413k	414k	416k	418k
Andorra	2650	2650	2650	2650	2650	2650	2650	2650	2650	2650	2650
UAE	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k
Argentina	534k	520k	506k	492k	479k	466k	453k	441k	429k	417k	420k
Armenia	413k	413k	413k	413k	413k	413k	413k	413k	413k	413k	413k
Antigua and Barbuda	37k	37k	37k	37k	37k	37k	37k	37k	37k	37k	37k
Australia	200k	205k	211k	216k	222k	227k	233k	239k	246k	252k	259k
Austria	3M	3.02M	3.04M	3.05M	3.07M	3.09M	3.11M	3.12M	3.14M	3.16M	3.18M
Azerbaijan	880k	880k	880k	880k	880k	880k	880k	880k	880k	880k	880k
Burundi	899k	899k	899k	899k	899k	899k	899k	899k	899k	899k	899k
Belgium	3.25M	3.26M	3.27M	3.28M	3.29M	3.3M	3.3M	3.31M	3.32M	3.33M	3.34M

Reshaping data…

Wide format to long format

Long format to wide format

Combining Datasets

Overview

Overview

01 Combining Datasets

Combining Datasets

1

Crucial step to gain comprehensive insights

2

Involves varying formats, structures, and data quality

3

Provides a broader perspective for more informed decisions

Overview

02 Stacking Datasets

Stacking Datasets

- Combining datasets vertically:
  - Identical variables
  - Different observations
- Adds more observations for a comprehensive analysis

Example: Combining two sets of weather data from two continents' countries

Example of Stacking: Rainfall and Temperature Data

country	average_rainfall	average_temperature
Kenya	24.8	71.6
Tanzania	42.2	78.8
Uganda	47.2	71.6
Ethiopia	33.4	70.2

Average annual rainfall (inches) and temperature (°Fahrenheit) in four countries

country	average_rainfall	average_temperature
China	25.4	57.2
Japan	66.0	60.8
India	42.6	75.2

Average annual rainfall (inches) and temperature (°Fahrenheit) in three countries

Example of Stacking: Rainfall and Temperature Data

country	average_rainfall	average_temperature
Kenya	24.8	71.6
Tanzania	42.2	78.8
Uganda	47.2	71.6
Ethiopia	33.4	70.2

Average annual rainfall (inches) and temperature (°Fahrenheit) in four countries

Resulting Stacked Table: Average annual rainfall and temperature in seven countries

country	average_rainfall	average_temperature
China	25.4	57.2
Japan	66.0	60.8
India	42.6	75.2

Average annual rainfall (inches) and temperature (°Fahrenheit) in three countries

Overview

03 Augmenting Datasets

Augmenting Datasets

- Combining datasets horizontally
  - Different variables
  - Shared observations

- Adds more variables to the dataset

Example: Combining weather variables from two seasonal data sets

Example of Augmenting: Seasonal Data of Japan

season	average_rainfall
Fall	16.5
Winter	16.7
Spring	9.7
Summer	23.1

Japan's average rain (inches) for four seasons

season	average_temperature
Fall	59.4
Winter	43.2
Spring	58.6
Summer	81.9

Japan's average temperature (°Fahrenheit) for four seasons

Example of Augmenting: Seasonal Data of Japan

season	average_rainfall
Fall	16.5
Winter	16.7
Spring	9.7
Summer	23.1

Japan's average rain (inches) for four seasons

season	average_temperature
Fall	59.4
Winter	43.2
Spring	58.6
Summer	81.9

Japan's average temperature (°Fahrenheit) for four seasons

Resulting Augmented Table:

Japan's average rain and temperature for four seasons

Overview

04 Merging Datasets

Merging Datasets

- Combining datasets horizontally and vertically
    - May involve different variables
    - May involve different observations
    - Observations matched on a common variable
- Adds more variables **and** observations to the dataset

Example of Merging: Student Life Datasets

student_id	club_name
S1	Drama Club
S1	Math Club
S2	Debate Team
S3	Math Club
S3	Science Society
S4	Drama Club
S5	Debate Team
S5	Science Society

Club membership data

student_id	major	gpa
S1	Math	3.8
S2	English	3.5

Academic performance data.

Example of Merging: Student Life Datasets

student_id	club_name
S1	Drama Club
S1	Math Club
S2	Debate Team
S3	Math Club
S3	Science Society
S4	Drama Club
S5	Debate Team
S5	Science Society

Club membership data

student_id	major	gpa
S1	Math	3.8
S2	English	3.5

Academic performance data.

Example of Merging: Student Life Datasets

student_id	club_name		
S1	Drama Club	Math	3.8
S1	Math Club		
S2	Debate Team		
S3	Math Club		
S3	Science Society		
S4	Drama Club		
S5	Debate Team		
S5	Science Society		

Club membership data.

student_id	major	gpa
	Math	3.8
S2	English	3.5

Academic performance data.

Club Membership and Academic Performance

Overview

05 Ethics in Practice: Avoiding Othering

Pearson

Ethics in Practice: Avoiding Othering

- Consider the impact on perception and understanding of personal identity attributes when grouping individuals under 'other'

- Ensure data integrity by avoiding the practice of 'othering' and by promoting diversity and inclusiveness

Summary

01 Combining Datasets

02 Stacking Datasets

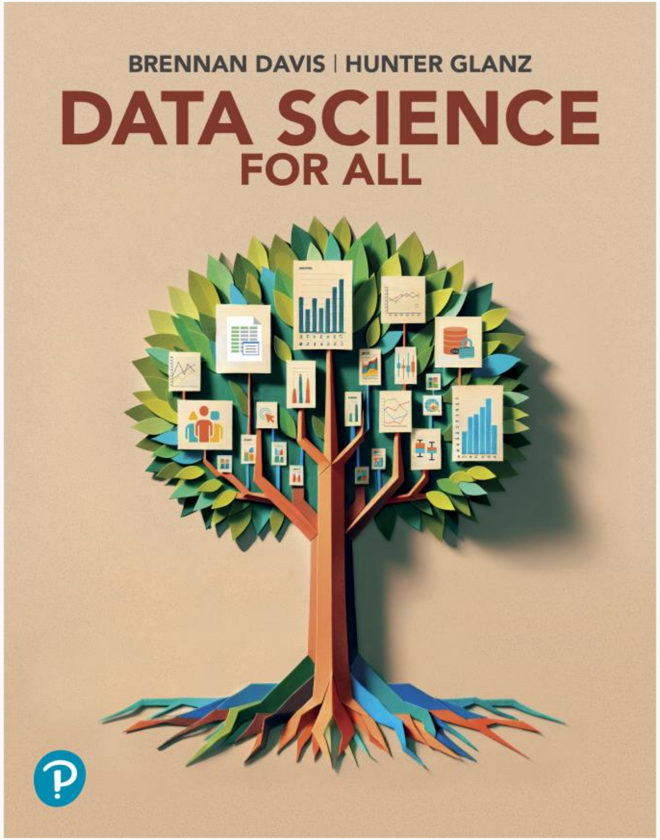
03 Augmenting Datasets

04 Merging Datasets

05 Ethics in Practice: Avoiding Othering

Chapter 3
Making Sense of Data Through Visualization

Visualizations with One Quantitative Variable



Overview

Visualizations with One Quantitative Variable

01 Histograms

02 Density Plots

03 Box Plots

04 Violin Plots

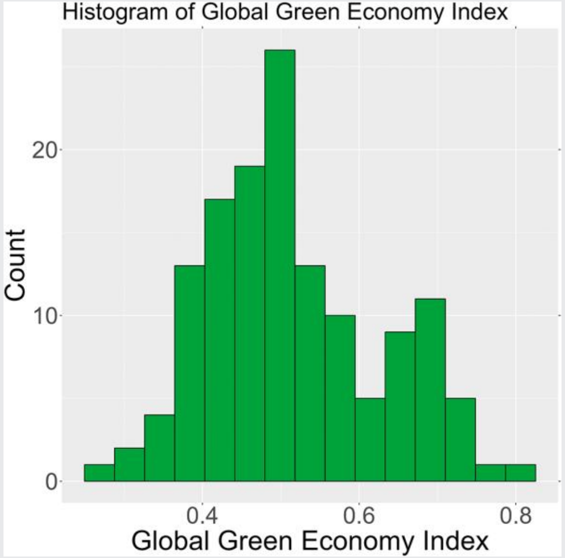
Pearson

Overview

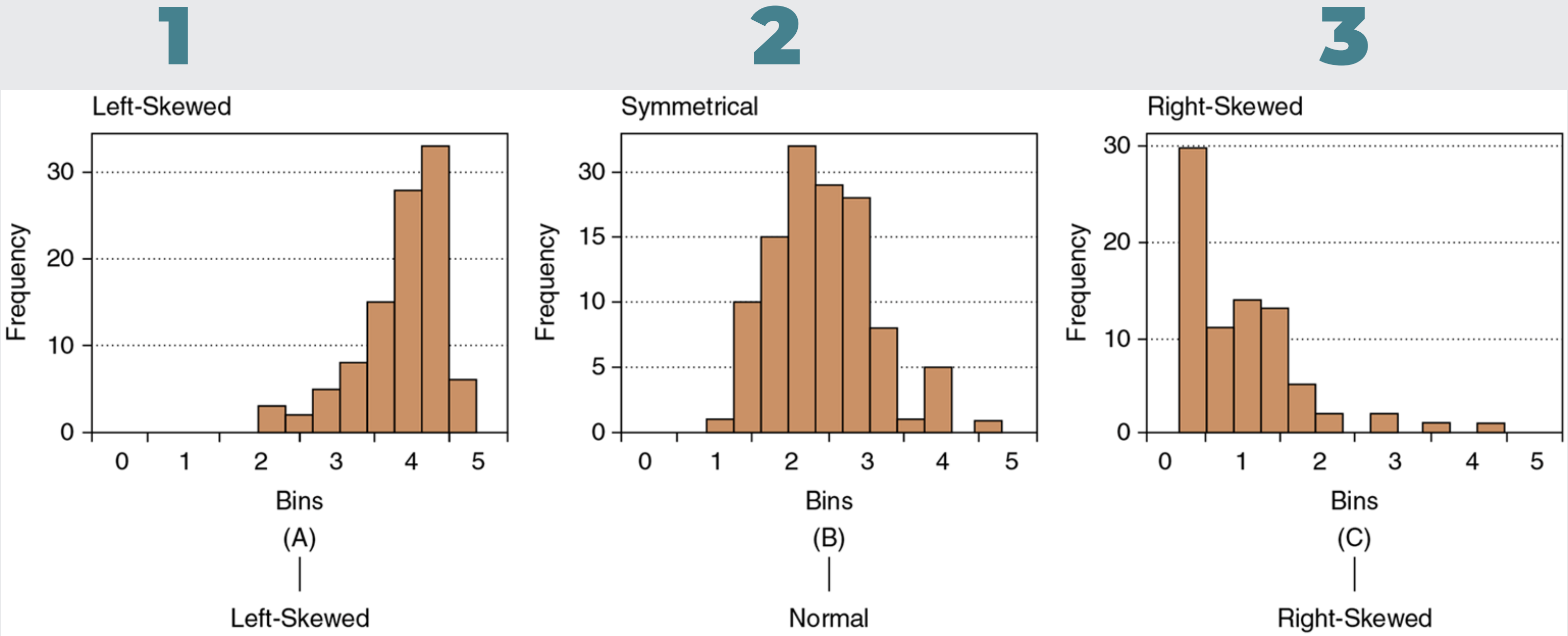
01 **Histograms**

Histograms

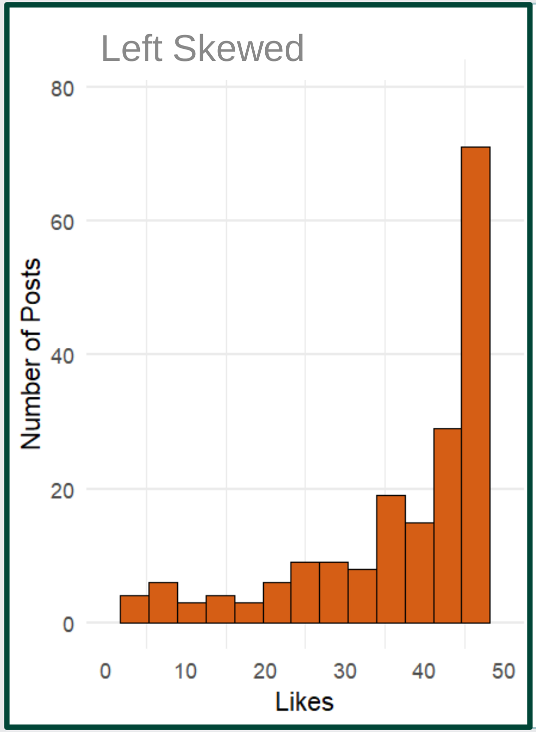
- Display the distribution of continuous data
- Show frequency of data values within intervals
- Vary in shape
    - Symmetrical
    - Left-skewed
    - Right-skewed

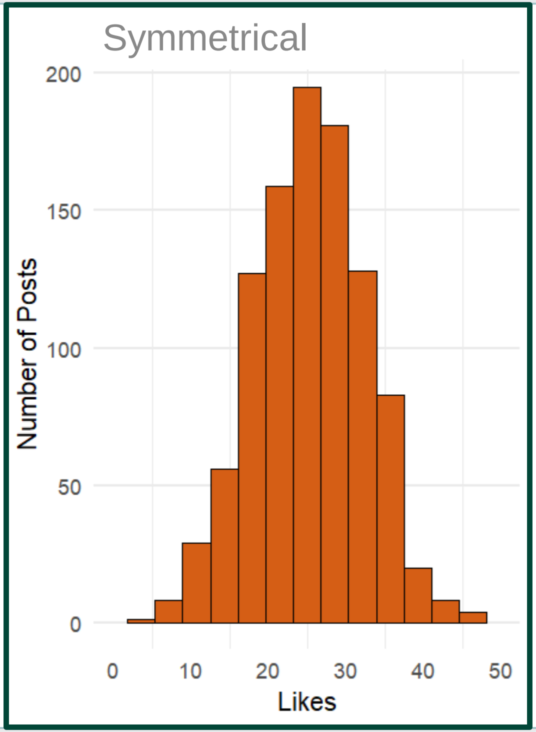


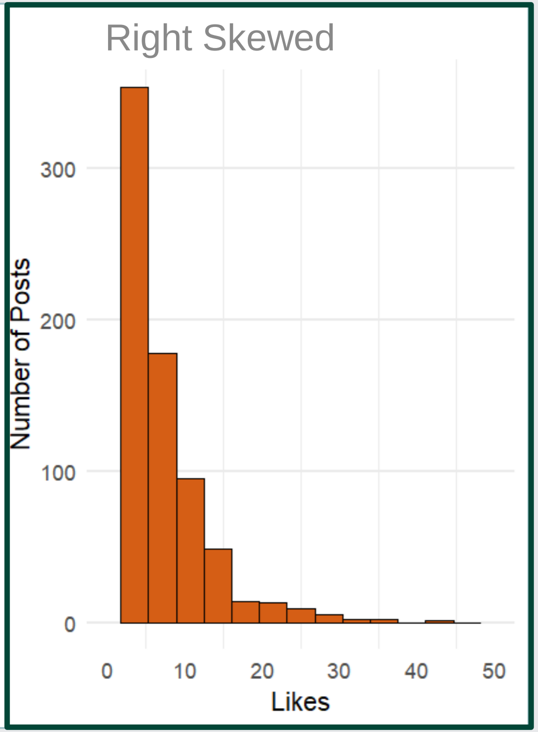
Histogram Symmetry and Skewness



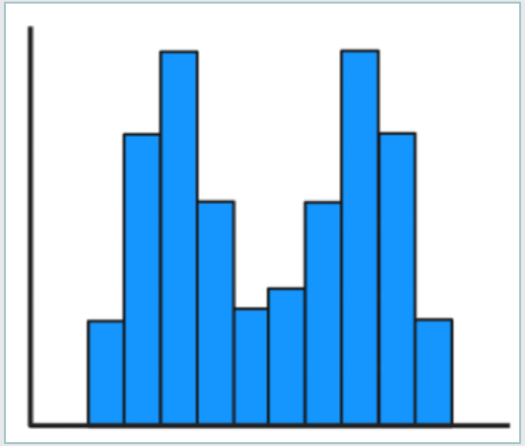
Applying Histograms to Social Media Analytics





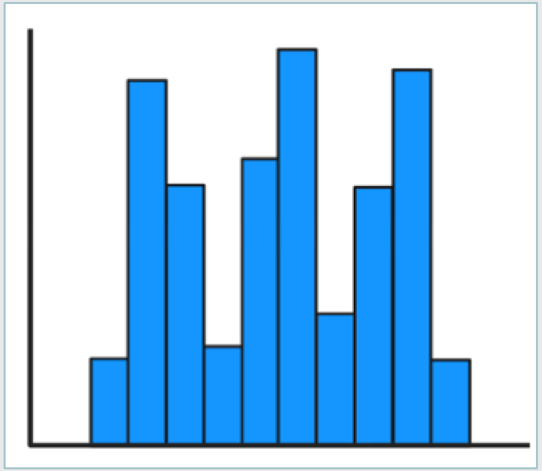


Bimodal and Multimodal Histograms



Bimodal Histogram

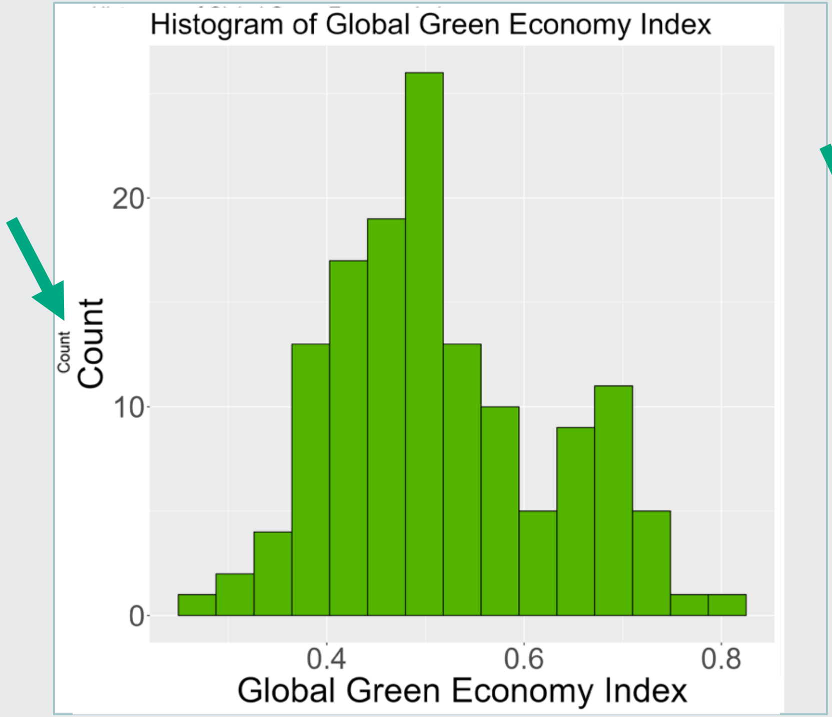
Two distinct peaks
Two different modes

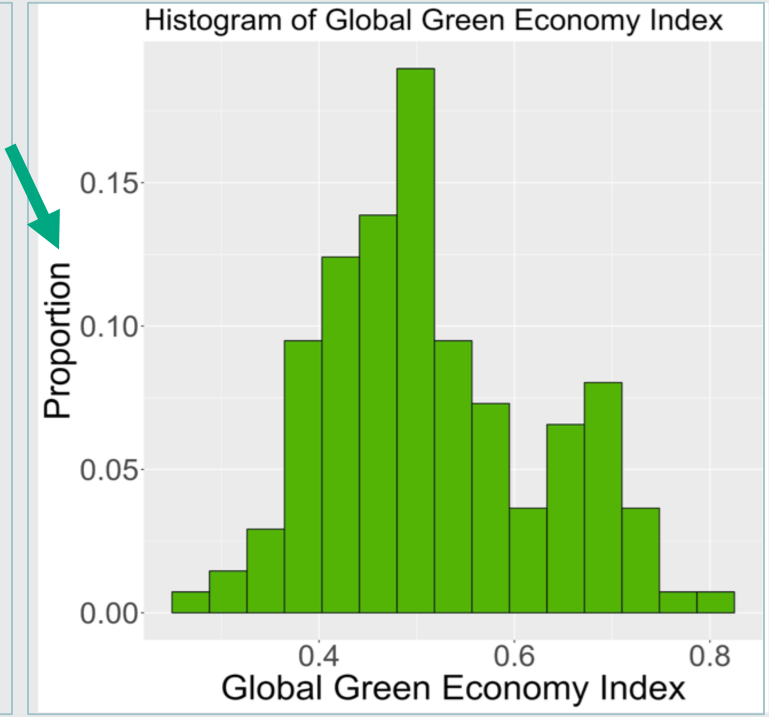


Multimodal Histogram

More than two peaks
Multiple modes

Exploring the Global Green Economy Index (GGEI) with Histograms

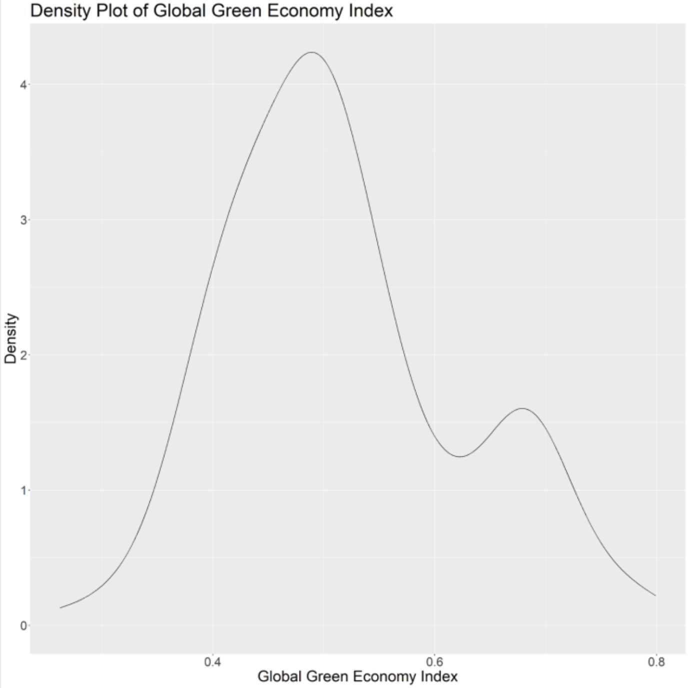




Overview

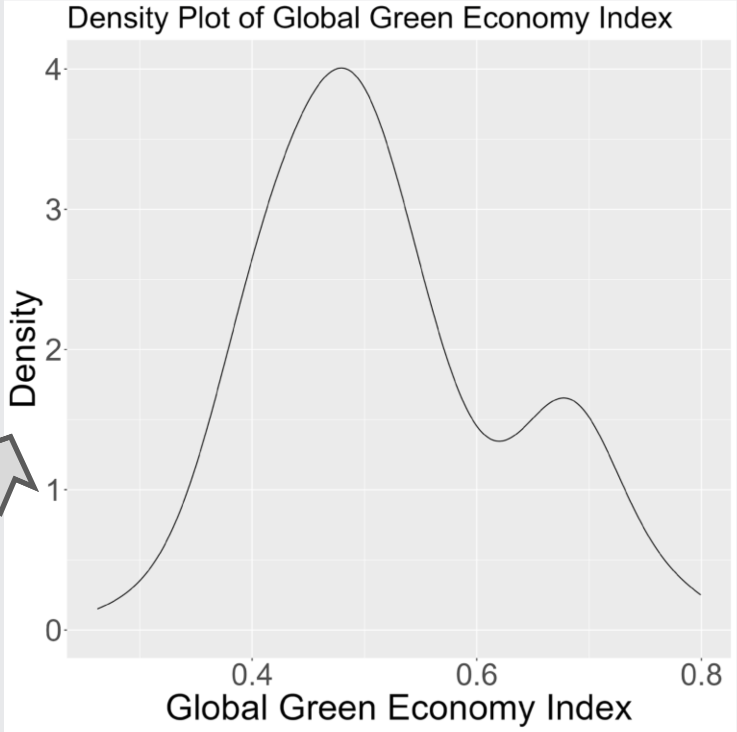
02 Density Plots

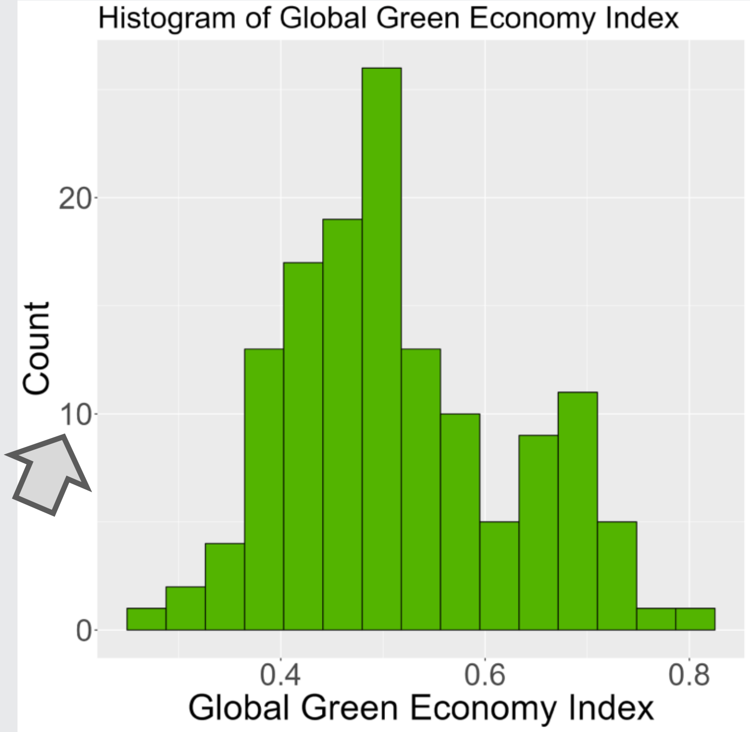
Density Plots



- Smoothed, continuous representation
- Identifies patterns and trends
- Enhances visual appeal

Comparing Histograms and Density Plots

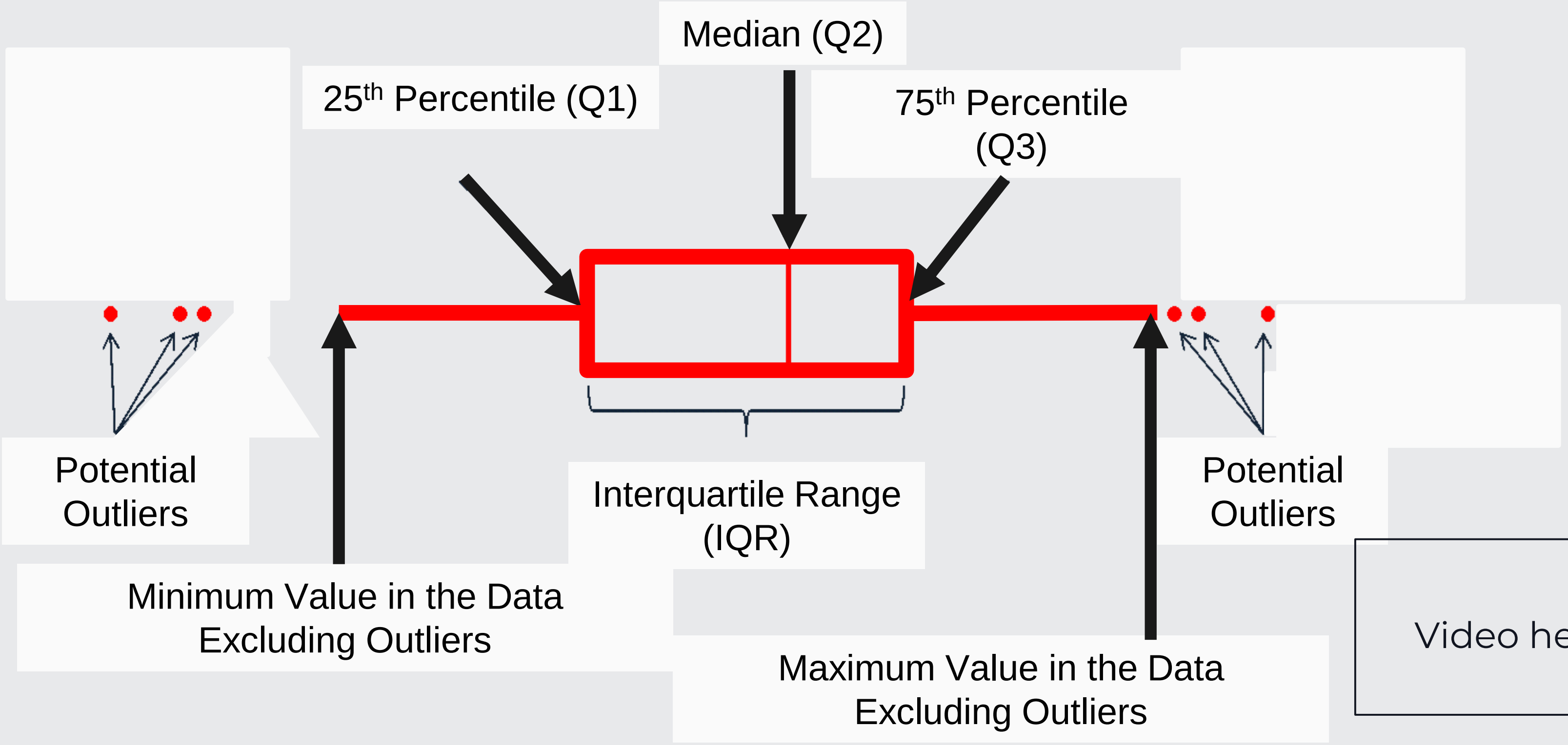




Overview

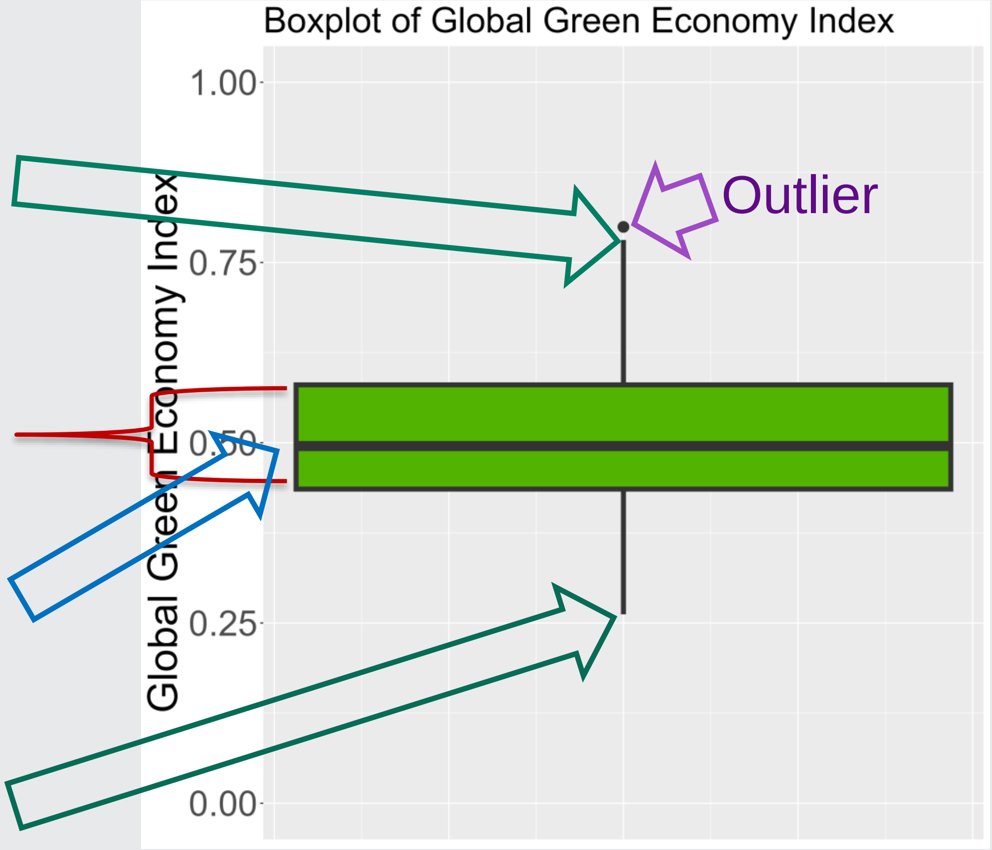
03 **Box Plots**

Box Plots



Video here

Box Plot Example



Maximum value whisker

Middle 50% of the Data

Median

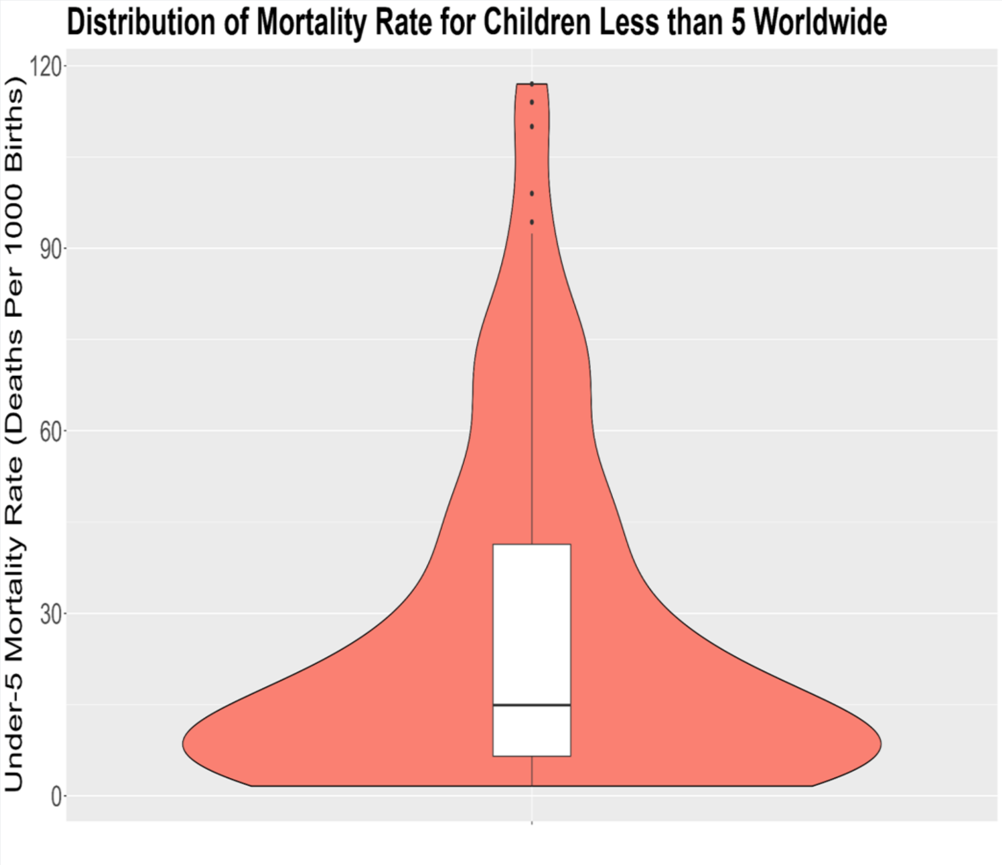
Minimum value whisker

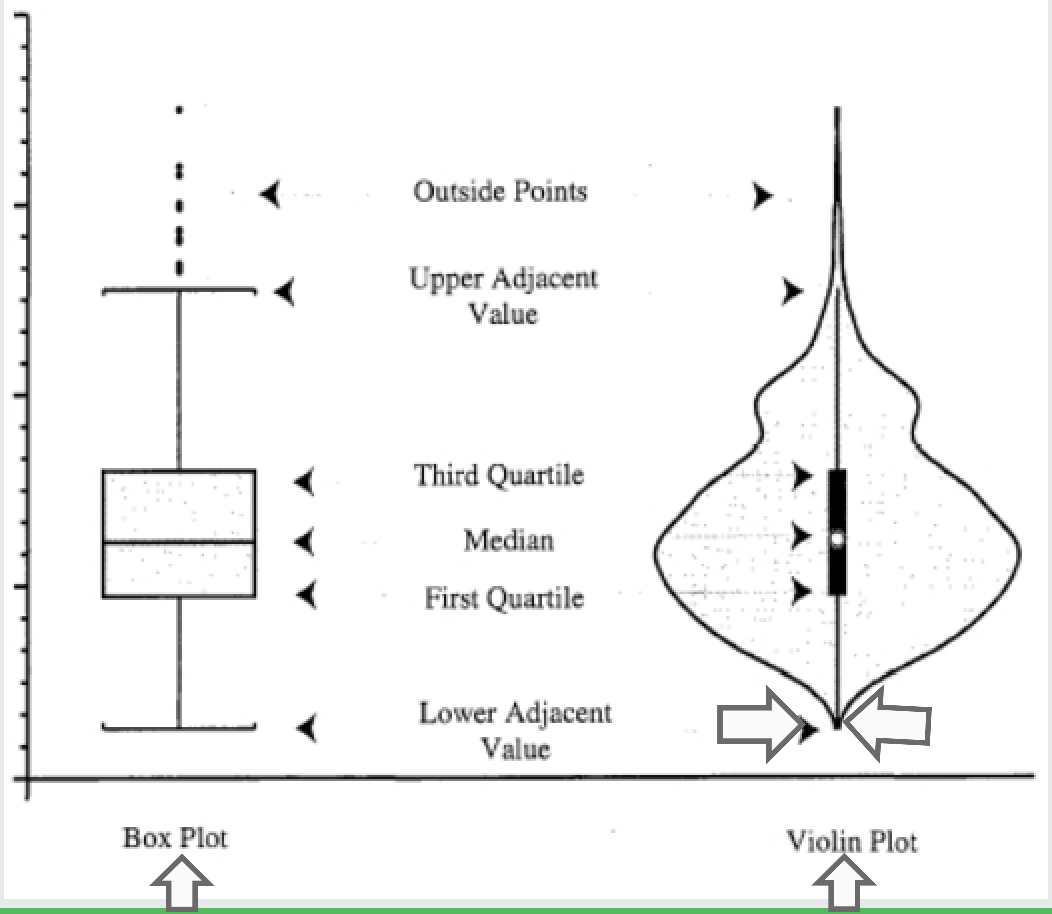
Overview

04 Violin Plots

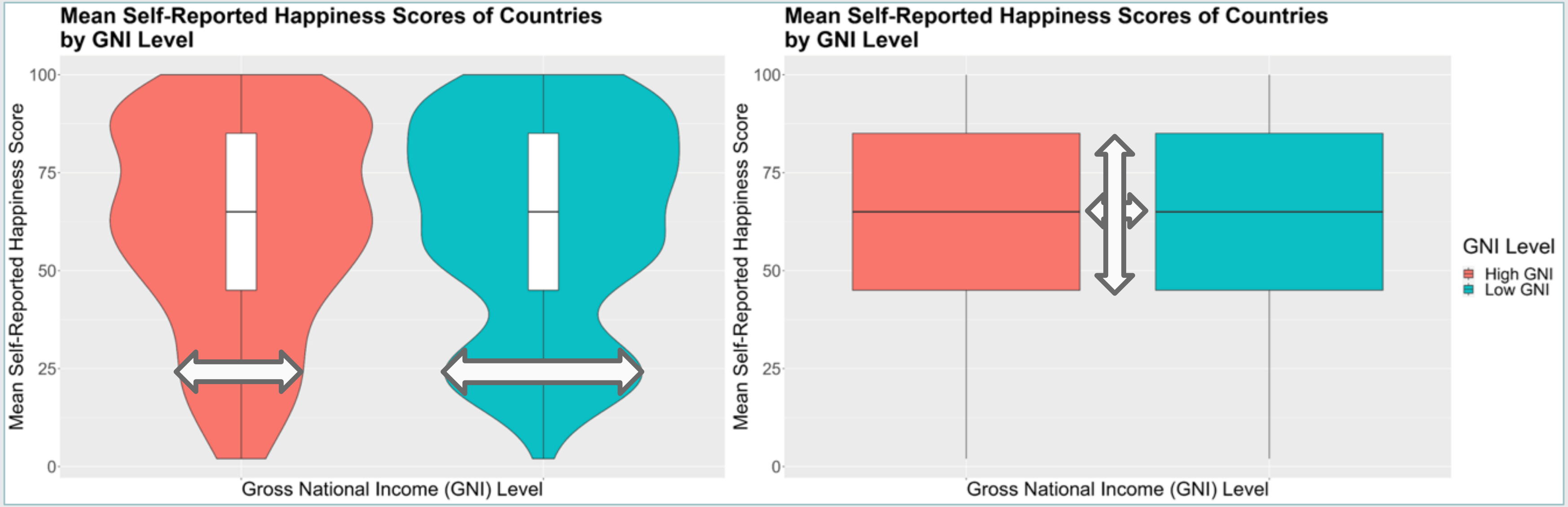
Violin Plots

- Combine features of box plots and density plots
- Provide a more detailed view of data distribution



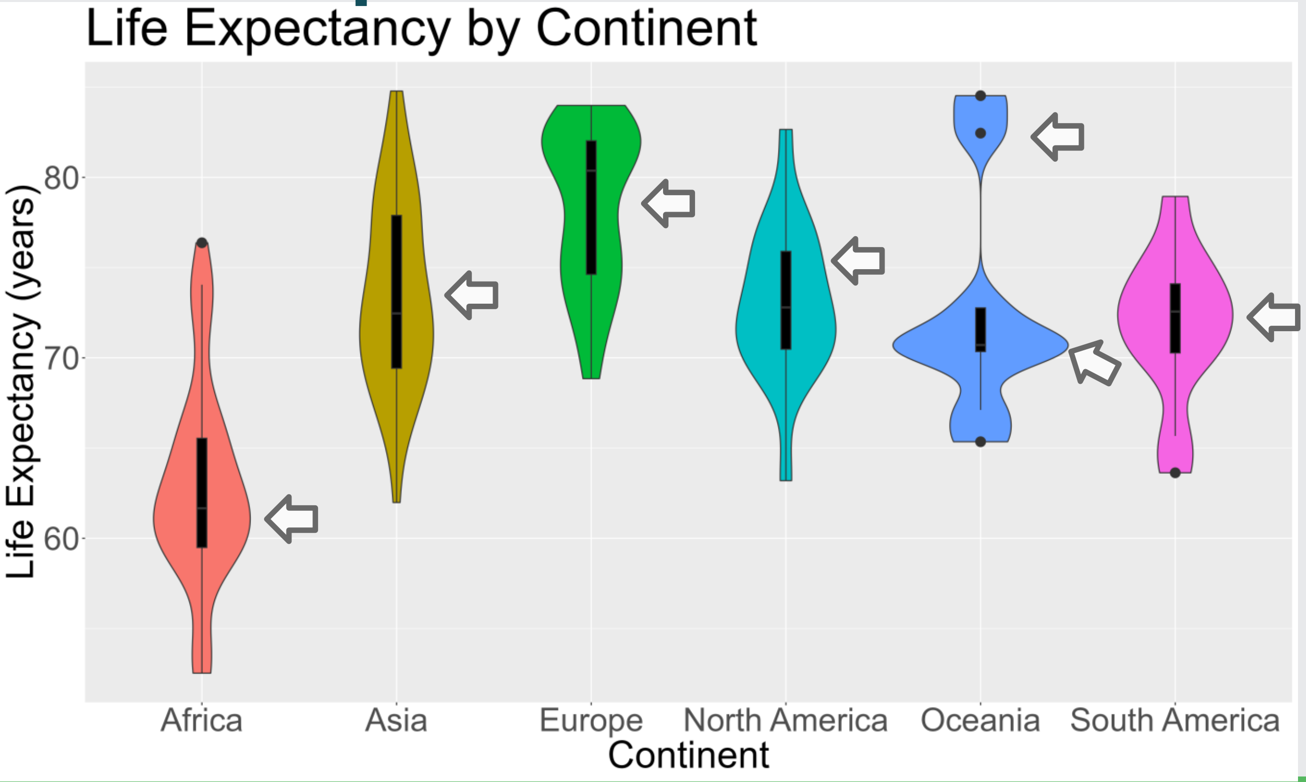

Outside Points
Upper Adjacent Value
Third Quartile
Median
First Quartile
Lower Adjacent Value
Box Plot
Violin Plot

Understand a Violin Plot versus Box Plot



Pearson

The Shapes of Violin Plots Differ



Summary

01 Histograms

02 Density Plots

03 Box Plots

04 Violin Plots

Overview

Visualization with One Categorical Variable

01 Bar Charts

02 Word Clouds

Overview

01 **Bar Charts**

Categorical Variables

- Values that represent category or group membership, like **Favorite Pet** and **Voted**

  All text variables are categorical variables

  Some numeric variables are categorical variables

  Mathematical operations are never used on categorical variables

First Name	Age	GPA	Favorite Pet	Voted
Selma	18	3.7	Dog	0
Jerry	19	3.5	Cat	1
Sanjiv	17	3.1	Dog	1
Olivia	18	3.9	Other	0
Hector	18	3.6	Fish	0

Categorical Variables

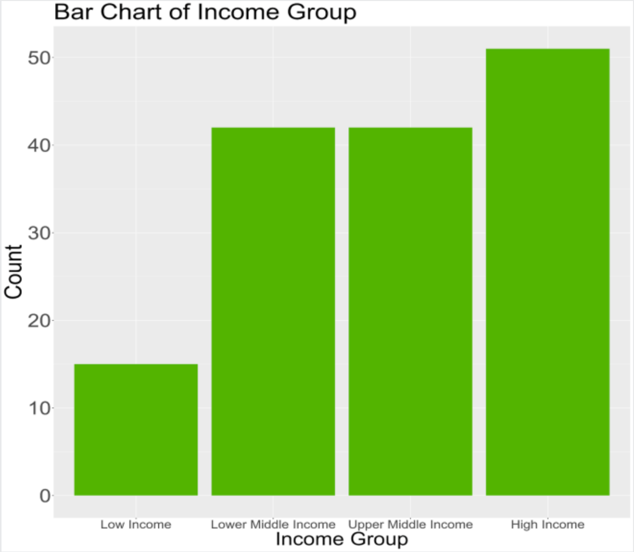
➡ Summarized as **counts** or **percentages**

For example…

- How many people responded "Dog" for Favorite Pet? "Cat"?
- What percentage of people indicated that they voted?

First Name	Age	GPA	Favorite Pet	Voted
Selma	18	3.7	Dog	0
Jerry	19	3.5	Cat	1
Sanjiv	17	3.1	Dog	1
Olivia	18	3.9	Other	0
Hector	18	3.6	Fish	0

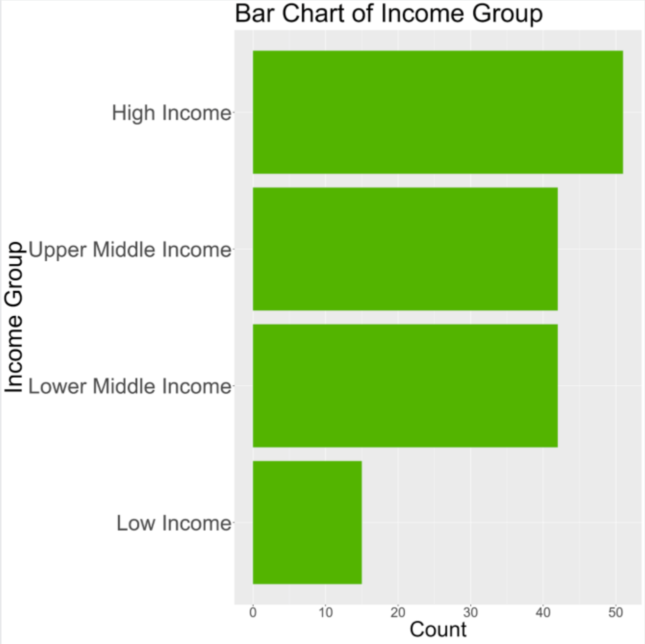
Column Charts



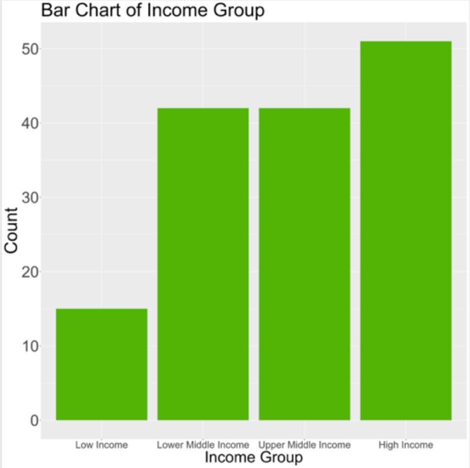
- Displays a frequency distribution for a categorical variable
- Shows the number of observations in each category of the variable

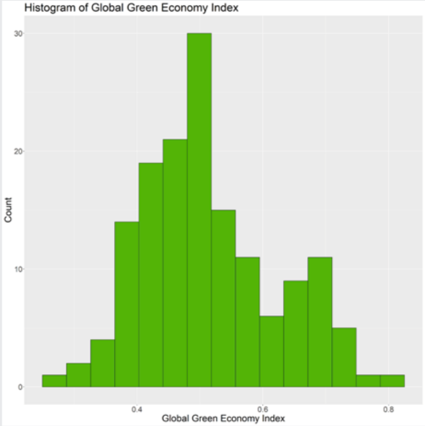
Bar Charts

- Column chart rotated so that the categorical variable is on the y-axis



Bar Charts vs. Histograms





Bar charts and **histograms** both have bars! Remember the differences!

- Column or bar charts for *categorical data*; Histograms for *quantitative data*
- Column or bar charts: space between bars. Histograms: adjacent bins

Overview

02 Word Clouds

Word Clouds

- Collection of words from a set of text data, shown in different sizes
- Words appearing more frequently in the data appear larger in the cloud

Summary

Bar Charts

Word Clouds

Overview

Visualizations with Two Variables

01 Bar Charts and Treemaps

02 Quantitative Variables

Overview

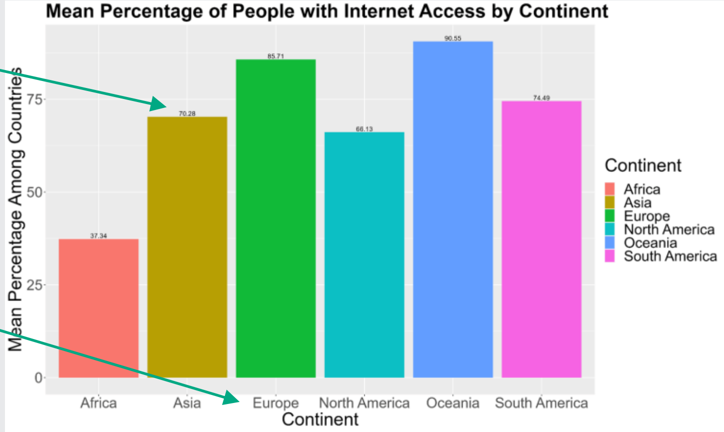
01 **Bar Charts and Treemaps**

Bar Charts for Quantitative and Categorical Variables

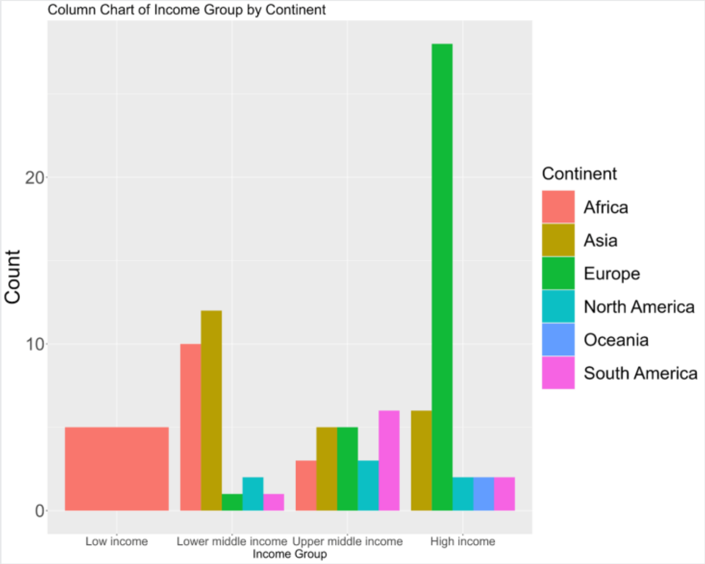
➡ Visualize the value of a quantitative variable for each value of a categorical variable

✦ Height of each bar = mean percentage of people with internet access (quantitative variable)

Each bar represents a continent (categorical variable)

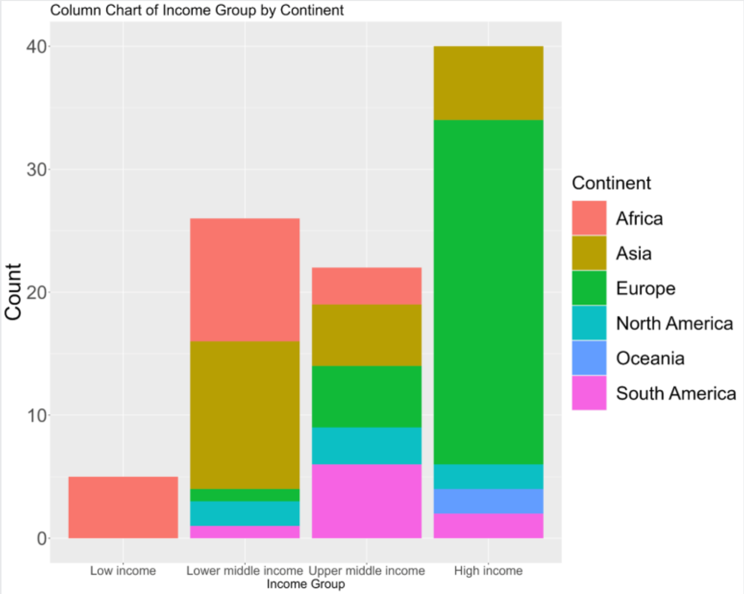


Bar Charts for Two Categorical Variables



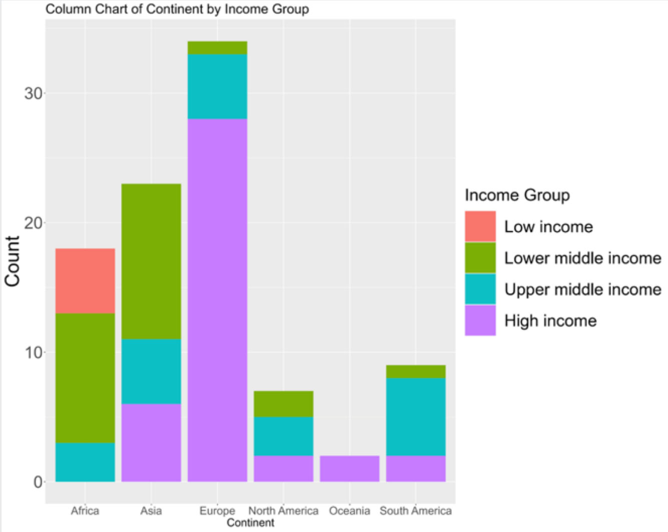
- Visualize two categorical variables in a **side-by-side** bar chart

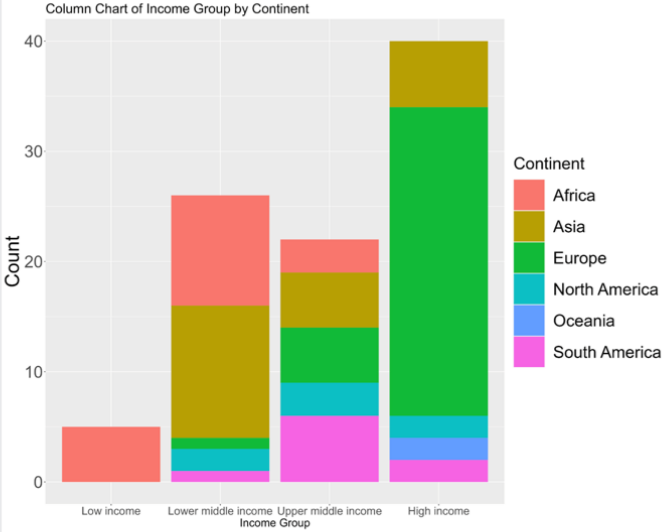
Bar Charts for Two Categorical Variables



- Visualize two categorical variables together in a **stacked** bar chart
- We have the same information in the plot here as we did in the side-by-side bar chart, but we call this a **stacked bar chart**

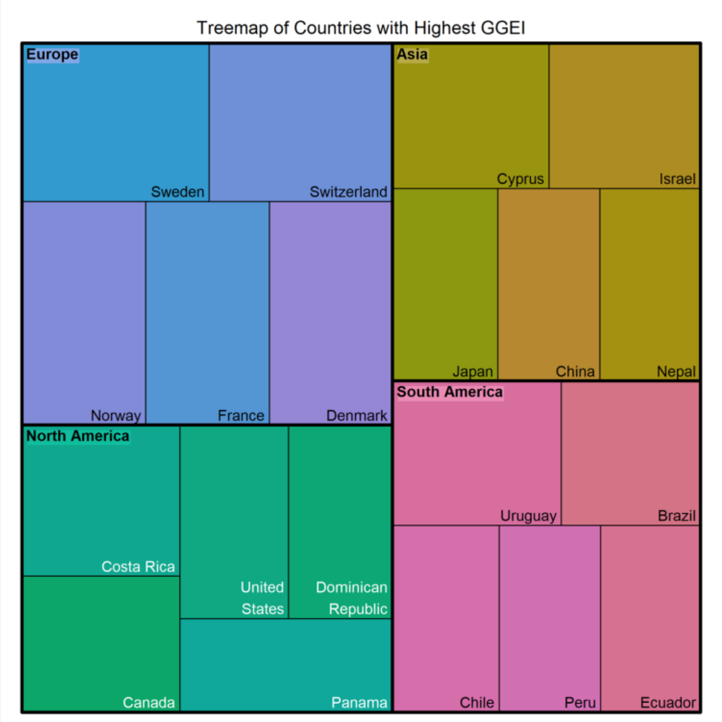
Bar Charts for Two Categorical Variables: Reversing the Variables





These plots display the exact same information for **Income Group** and **Continent**, but organized in different ways!

Treemaps



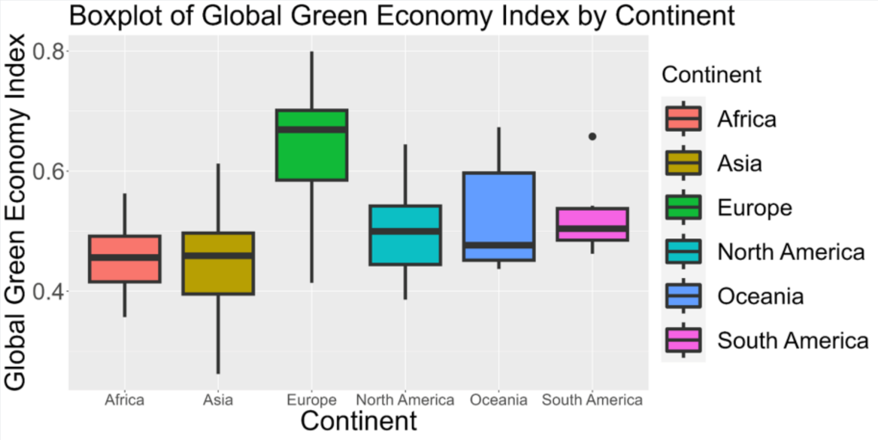
- Displays **hierarchical data** (e.g. country within continent)
- Each rectangle represents a category or subcategory
- Size of each rectangle corresponds to the value of the data it represents

Overview

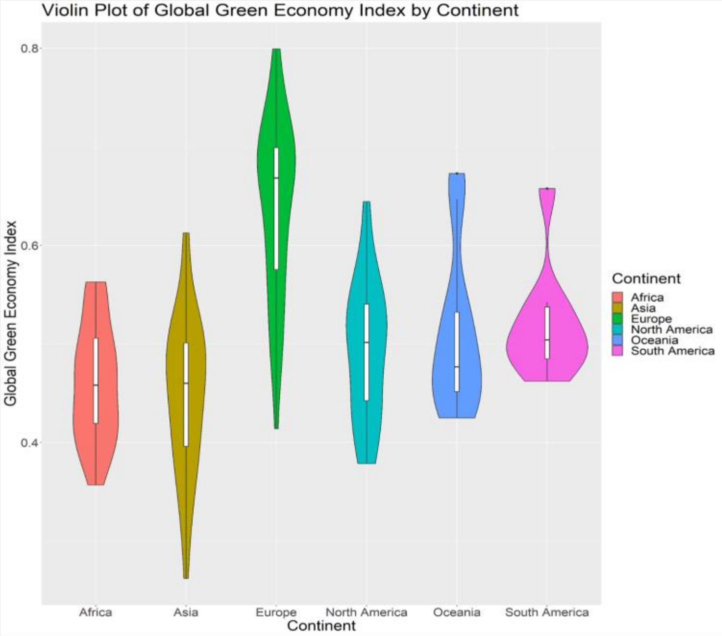
02 **Quantitative Variables**

Box Plot with a Categorical Variable

- Visualize a quantitative variable's distribution across the values of a categorical variable
- Box plots for the same variable grouped by a categorical variable

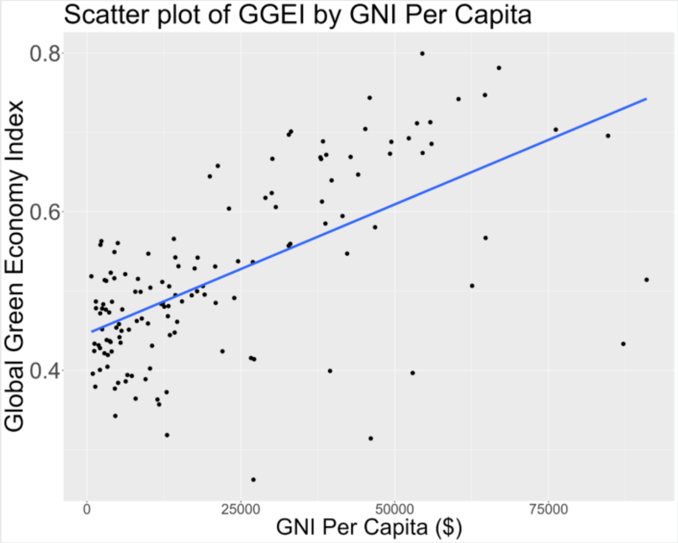


Violin Plot **with a Categorical Variable**



- Visualize a quantitative variable's distribution across the values of a categorical variable
- Violin plots for the same variable grouped by a categorical variable

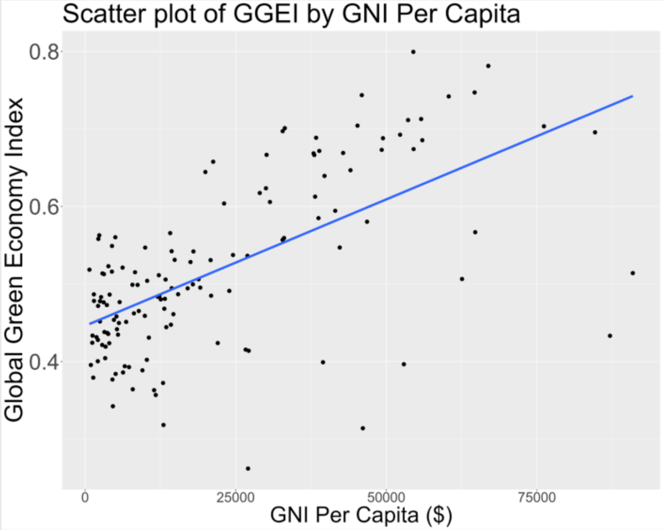
Scatterplot



- Displays the relationship between two quantitative variables
- Each point represents the values of the two variables

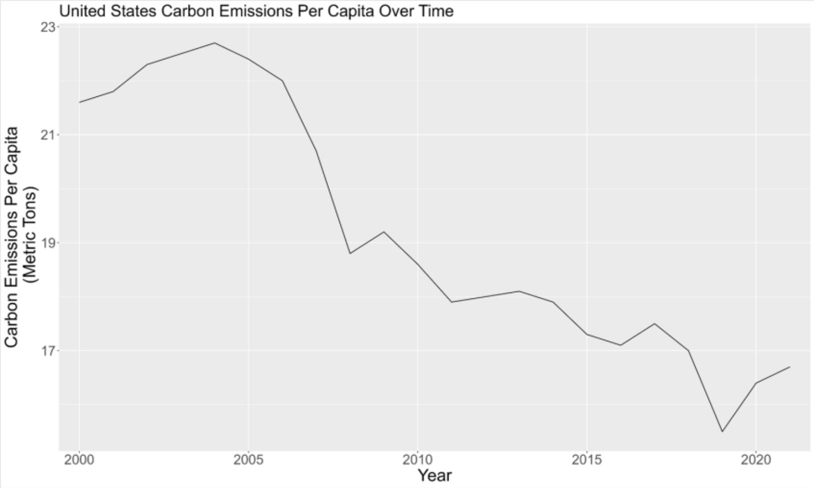
Pearson

Explanatory and Response Variables



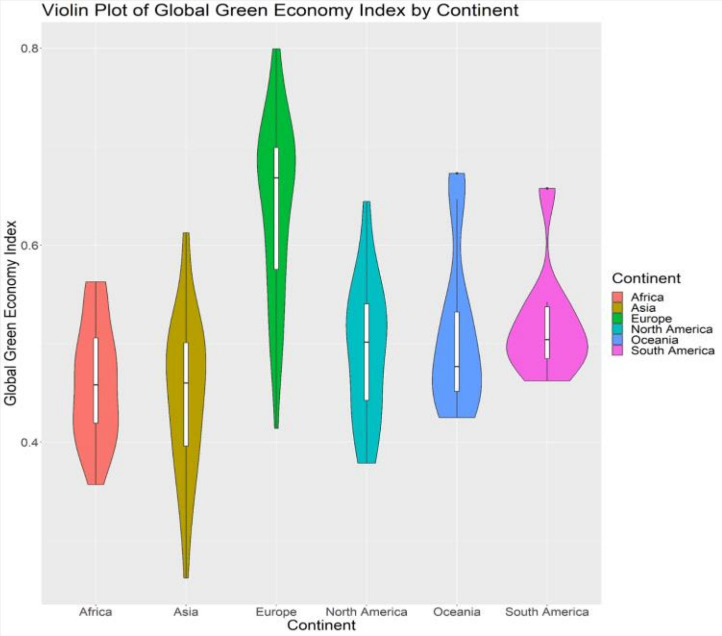
- **Response variable:** Primary variable of interest in our data set. Thought to be affected by, influenced by, or associated with one or more explanatory variables.
  - Plotted on the y-axis
- **Explanatory variable:** Thought to affect, influence, explain, or be associated with the response variable.
  - Plotted on the x-axis

Line Chart



- Displays data as a series of points connected by lines, typically showing a value over time

Summary



Bar Charts and Treemaps

Quantitative Variables

Overview

Visualizations with Three or More Variables

01 Scatterplots

02 Bubble Charts

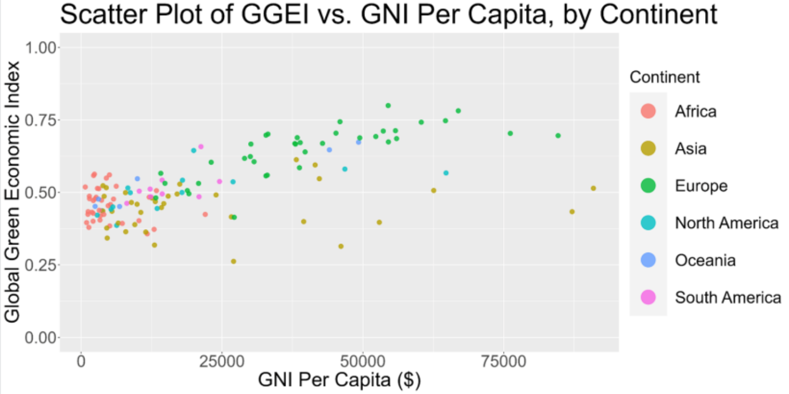
03 Heatmaps

04 Geographic Maps

Overview

01 Scatterplots

Visualize Two Quantitative Variables and One Categorical Variable with a Scatterplot



- Two quantitative variables on the x- and y-axes, like a scatterplot
- Color points, to indicate values of a third, categorical variable

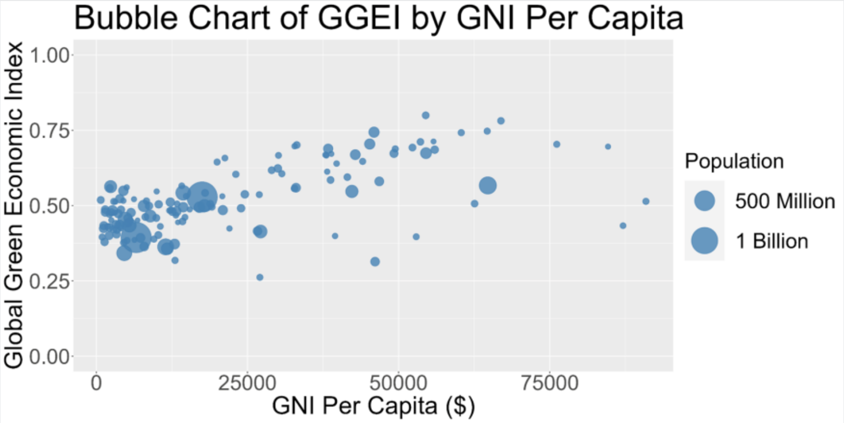
Overview

02 **Bubble Charts**

Bubble Charts for Three Quantitative Variables

➡ Displays three quantitative variables in a visually appealing way

- Two quantitative variables on the x- and y-axes, like a scatterplot
- Size the points by a third, quantitative variable

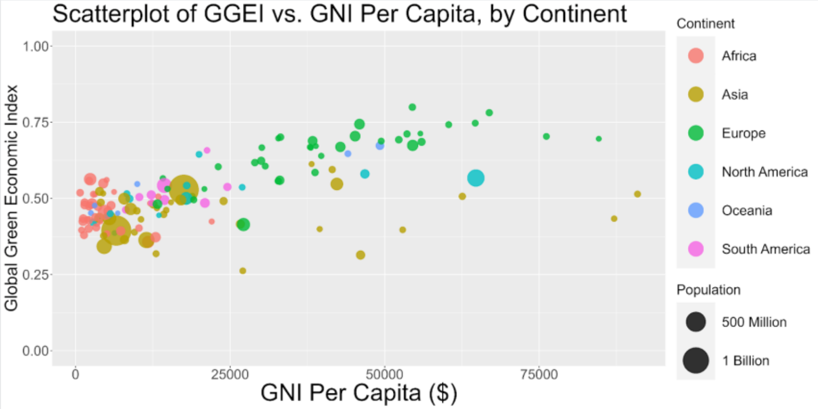


Pearson

Bubble Charts for Quantitative and Categorical Variables

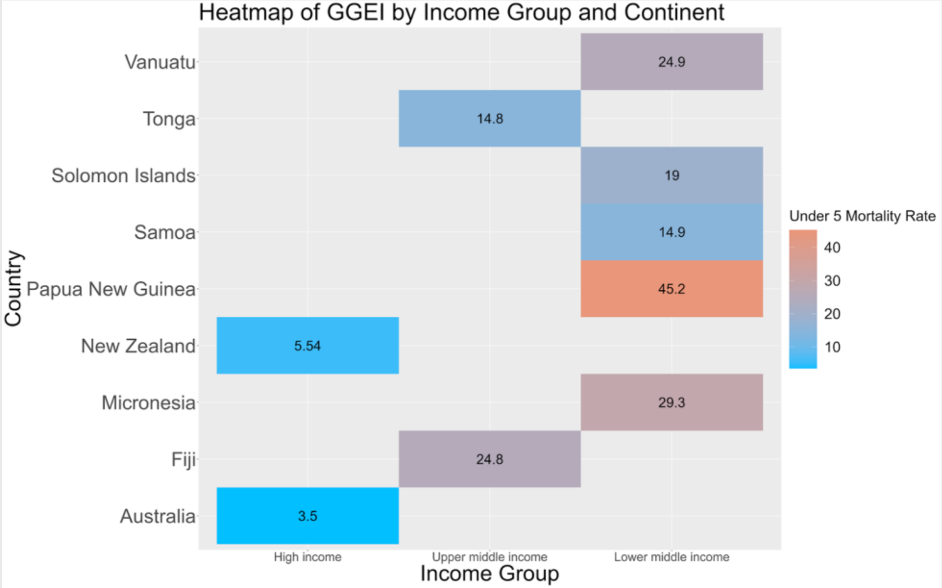
➡ Add a categorical variable to a traditional bubble plot using color

- Two quantitative variables on the x- and y-axes
- Color the points to indicate values of a categorical variable
- Size the points by values of an additional quantitative variable



Overview

03 **Heatmaps**



Heatmaps

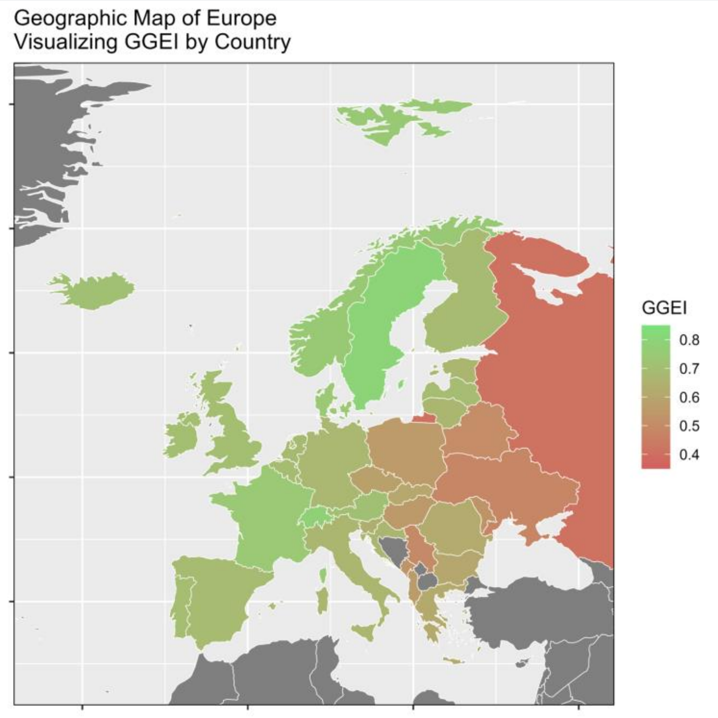
- ➡ Use color to represent quantitative data, for combinations of values of two categorical variables

Overview

04 Geographic Maps

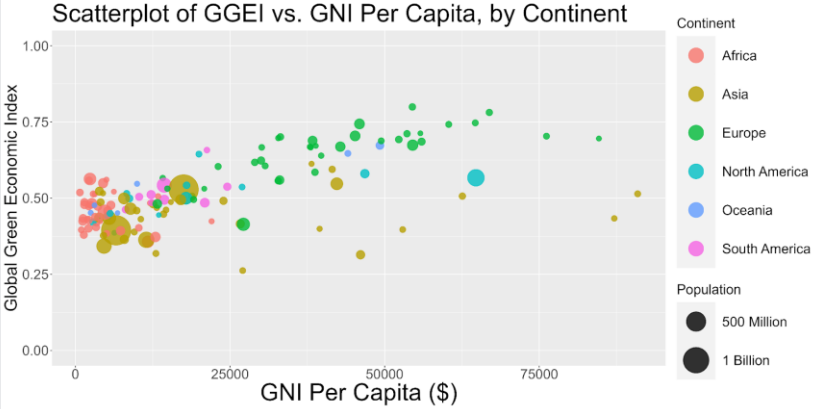
Geographic Maps

- Visualize a quantitative variable associated with geographic regions

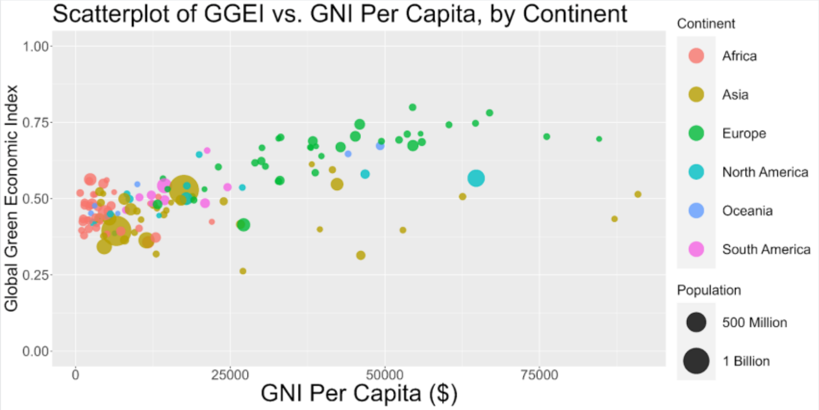


Big Picture Break: **Visualizing Data**

- The options for visualizing a particular variable or combination of variables are near infinite
- We've introduced many popular choices, but it's not uncommon to explore many options
- Data visualizations should be clear and accessible to anyone, and quickly digestible



Summary



Scatterplots

Bubble Charts

Heatmaps

Geographic Maps

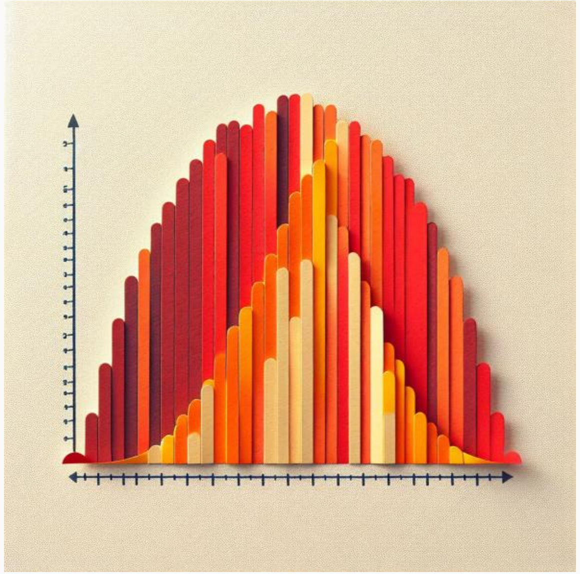
Central Tendency

Overview

01 Chapter 4 Vignette

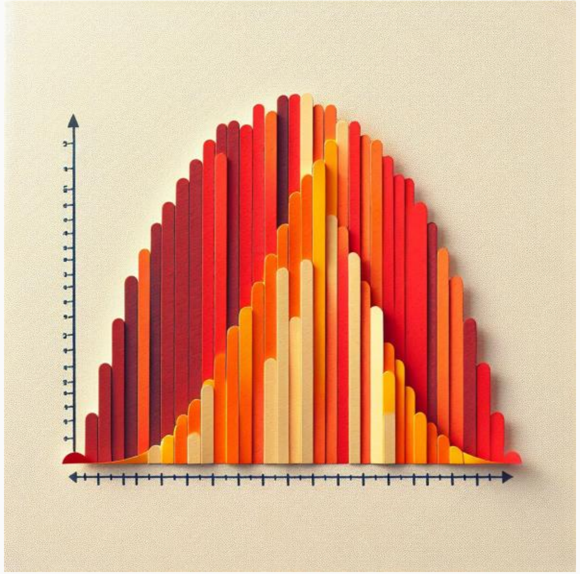
02 Case Study: Shopify

03 Central Tendency



Overview

01 Chapter 4 Vignette



Chapter 4 Vignette

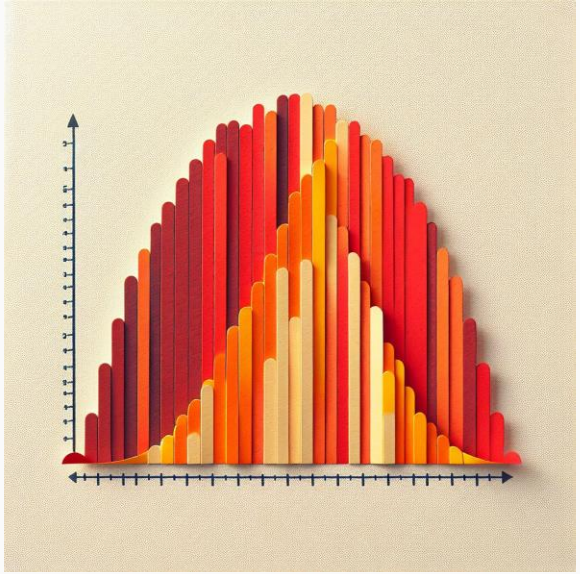
- Sam focused on saving up to buy his first car
- Parents offered to match his savings
- He needs to navigate market of pre-owned cars
- Devotes every spare moment to pursuit of the perfect used car
- Data set with 426,880 entries
- Exploratory data analysis topics: central tendency, variability, shape, associations, outliers

Chapter 4 Vignette

id	price	year	manufacturer	model	condition	cylinders	fuel	odometer	transmission	type	state
7316814884	33590	2014	gmc	sierra 1500 crew cab slt	good	8 cylinders	gas	57923	other	pickup	al
7316814758	22590	2010	chevrolet	silverado 1500	good	8 cylinders	gas	71229	other	pickup	al
7316814989	39590	2020	chevrolet	silverado 1500 crew	good	8 cylinders	gas	19160	other	pickup	al
7316743432	30990	2017	toyota	tundra double cab sr	good	8 cylinders	gas	41124	other	pickup	al
7316356412	15000	2013	ford	f-150 xlt	excellent	6 cylinders	gas	128000	automatic	truck	al
7316343444	27990	2012	gmc	sierra 2500 hd extended cab	good	8 cylinders	gas	68696	other	pickup	al
7316304717	34590	2016	chevrolet	silverado 1500 double	good	6 cylinders	gas	29499	other	pickup	al
7316285779	35000	2019	toyota	tacoma	excellent	6 cylinders	gas	43000	automatic	truck	al
7316257769	29990	2016	chevrolet	colorado extended cab	good	6 cylinders	gas	17302	other	pickup	al
7316133914	38590	2011	chevrolet	corvette grand sport	good	8 cylinders	gas	30237	other	other	al
7316130053	4500	1992	jeep	cherokee	excellent	6 cylinders	gas	192000	automatic		al
7315816316	32990	2017	jeep	wrangler unlimited sport	good	6 cylinders	gas	30041	other	other	al
7315770394	24590	2017	chevrolet	silverado 1500 regular	good	6 cylinders	gas	40784	other	pickup	al

Overview

02 **Case Study: Shopify**



Case Study: Shopify Helps Small Businesses with Descriptive Analytics

Case Study: Shopify Helps Small Businesses with Descriptive Analytics

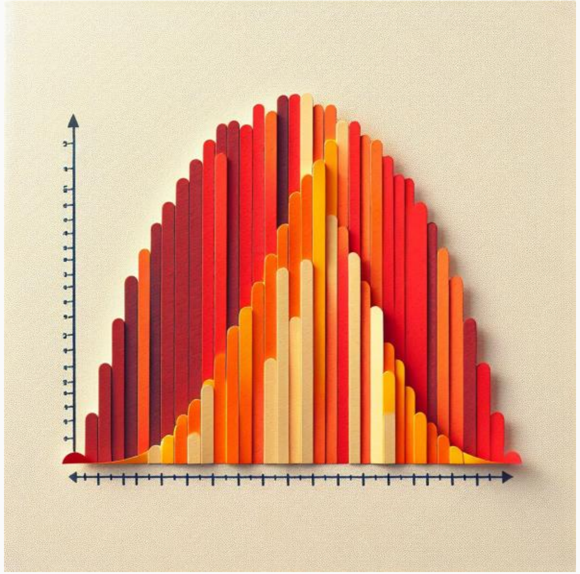
A = 80% of revenue

B = 15% of revenue

C = 5% of revenue

Overview

03 Central Tendency



Central Tendency

Definition

Central tendency characterizes a data set using the middle or "typical" value

Measures

Mean, Median, Mode, 90% Trimmed Average

Importance

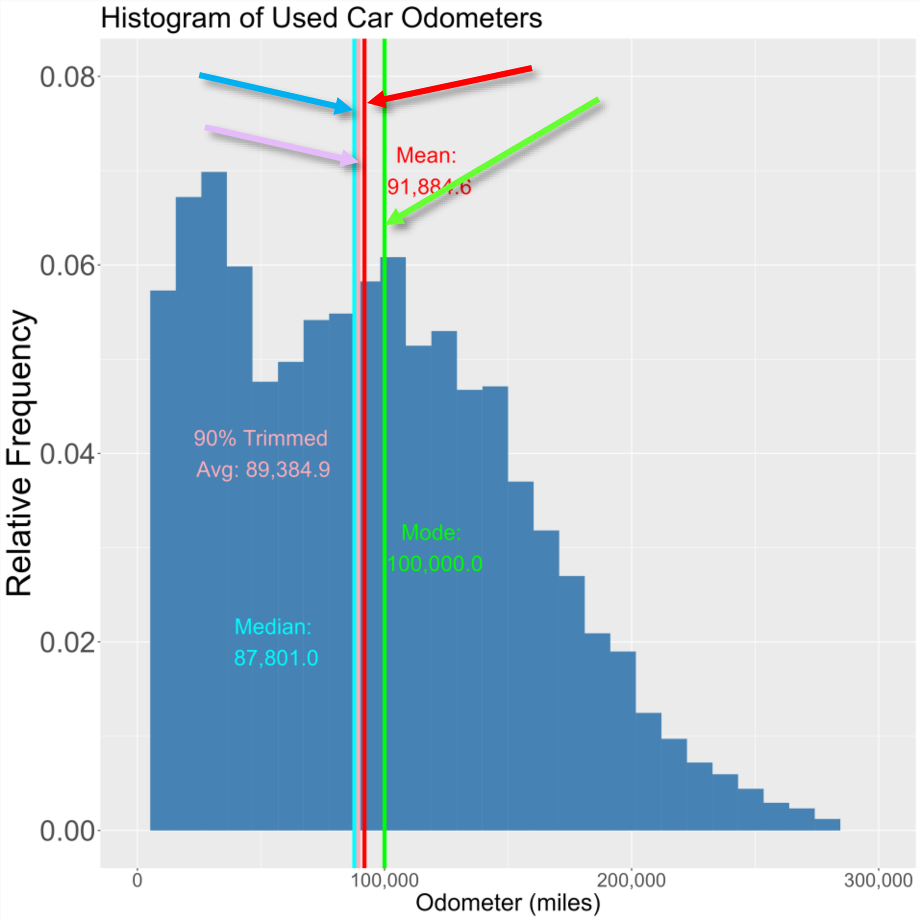
Summaries, comparisons, predictions

Central Tendency Definitions

- The **mean**, or average, is computed by adding all values in the data set and dividing by the number of observations.
- A **90% trimmed average** is the average of the middle 90% of a variable’s values in a data set.
- The **median** is the middle value in a data set
- The **mode** is the value that occurs the most frequently in the data set.

Central Tendency

- Mean: $26,533
- 90% trimmed average: $21,454
- Median: $20,000
- Mode: $25,000



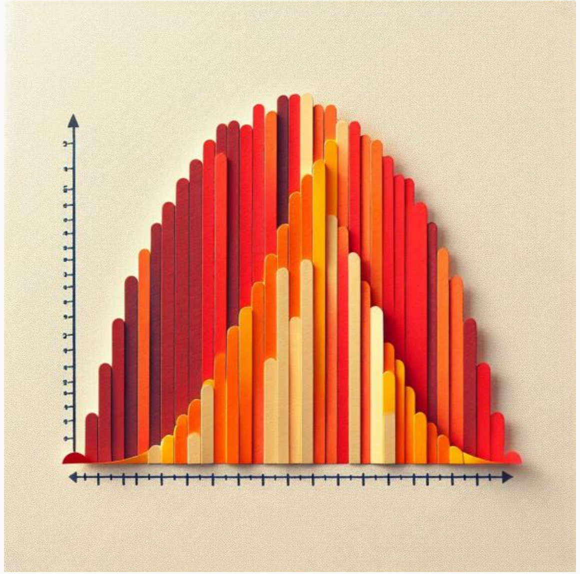
- Mean: 91,884.6 miles.
- Median: 87,801 miles.
- Mode: 100,000 miles.
- 90% trimmed average: 89,384.9

Summary

01 Chapter 4 Vignette

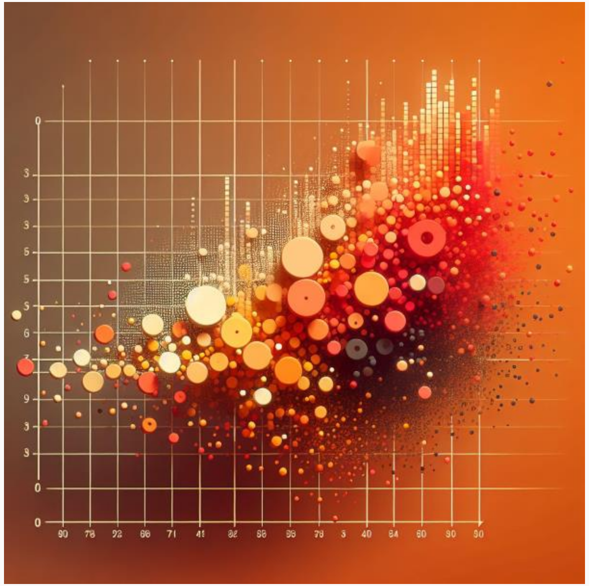
02 Case Study: Shopify

03 Central Tendency



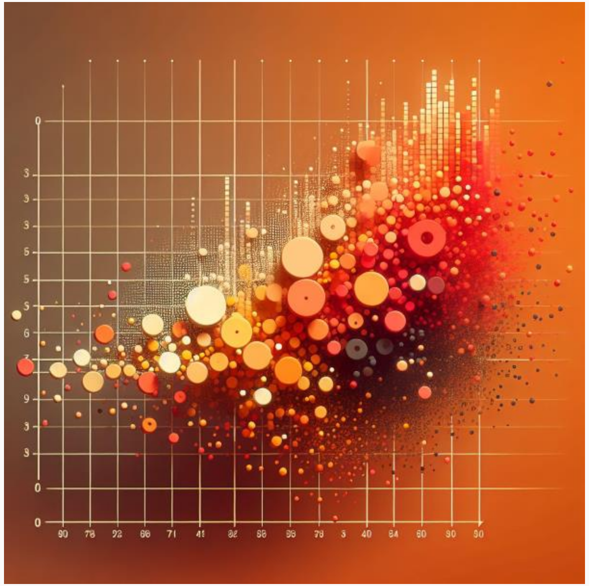
Overview

Variability



Overview

01 Variability



Variability

Measure of how much data points in a quantitative data set differ from each other

Measures of Variability

Range

Difference between maximum value and minimum value in a dataset, a simple measure of spread

Variance

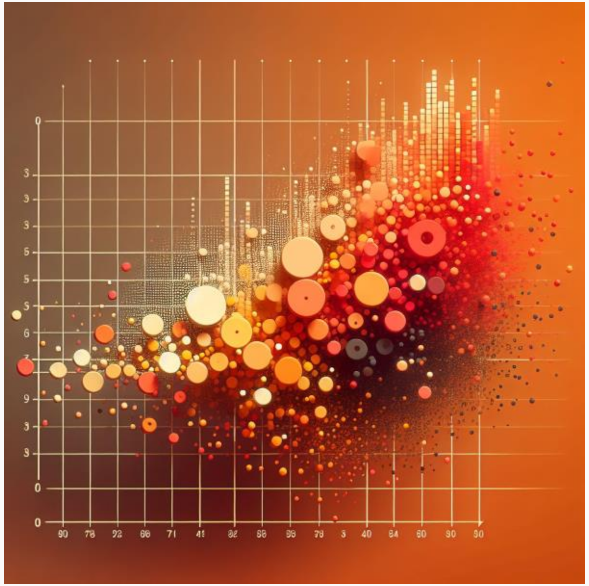
Average of the squared differences between data points and the mean, indicating the dispersion of values

Standard Deviation

Square root of the variance, a more interpretable measure of variability

Overview

02 Range, Variance, and Standard Deviation



Range

- Difference between the maximum and minimum values in a dataset

- Provides a simple measure of the spread of values

- **range** = **maximum value** – **minimum value**

Variance

- Average squared distance from each data point to the mean
- Quantifies how much data points differ from the average value

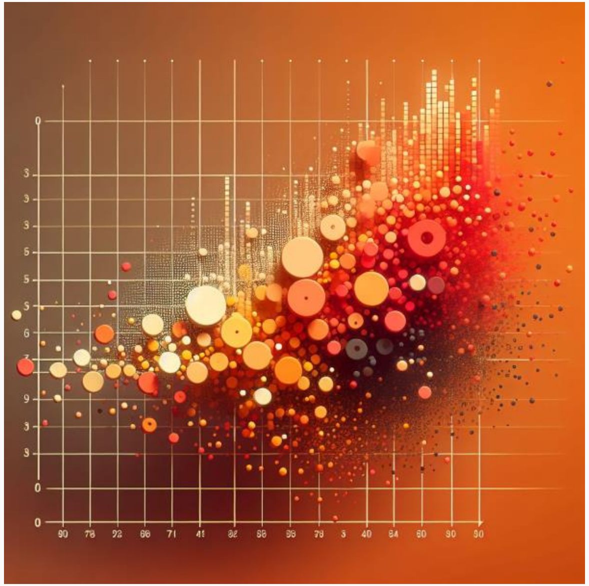
Standard Deviation

- Quantifies the amount of variation or dispersion in a set of values
- Calculated as the square root of the variance

- Smaller standard deviation indicates data points *closer* to the mean (*less* variability)
- Larger standard deviation indicates data points *further* from the mean (*greater* variability)

Overview

03 **Variability Example and Case Study**



Range Example

Range = **maximum value** – **minimum value**

Odometer Readings in Sam's data

- Highest = 281,700 miles
- Lowest = 0 miles
- Range = 281,700 – 0 = 281,700 miles

Variance Example

Sam asks his data science tool to calculate variance for the Odometer Readings in his data and gets the following response:

- 3,710,149,921 miles^2

Standad Deviation Example

Standard Deviation = $\sqrt{3{,}710{,}117{,}109}$ = 60,910.73, rounded to 60,911

One Standard Deviation *below* the mean = 91,885 – 60,911 = 30,974

One Standard Deviation *above* the mean = 91,885 + 60,911 = 152,796

		Odometer
Statistic	**Range**	281,700 miles
	Variance	3,710,149,921 miles^2
	Standard Deviation	60,911 miles

Case Study: Exploratory Data Analysis in Sports

Golden State Warriors

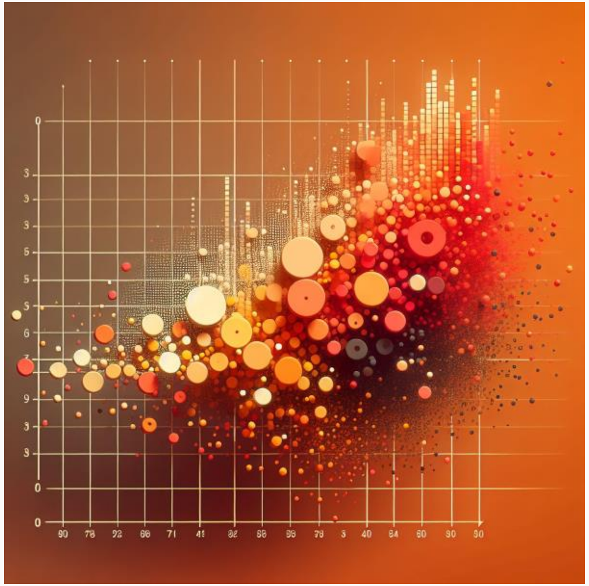
- Utilized data analysis to optimize player performance and game strategies
- Applied data-driven insights to help player recruitment, injury prevention, and decision-making during games

Summary

01 Variability

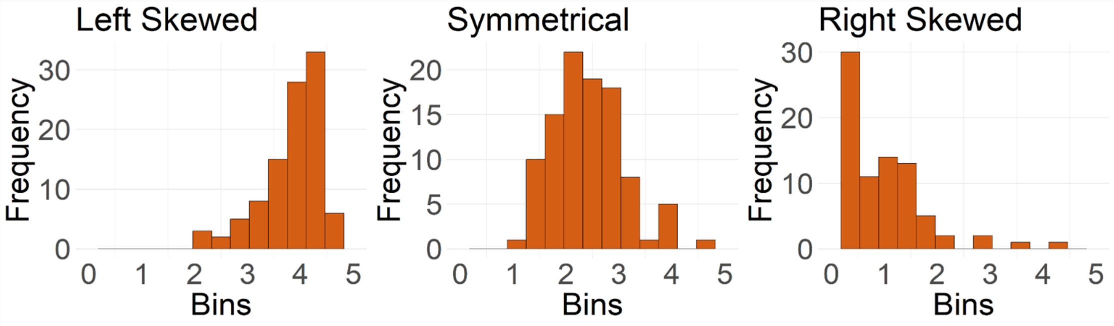
02 Range, Variance, and Standard Deviation

03 Variability Example and Case Study

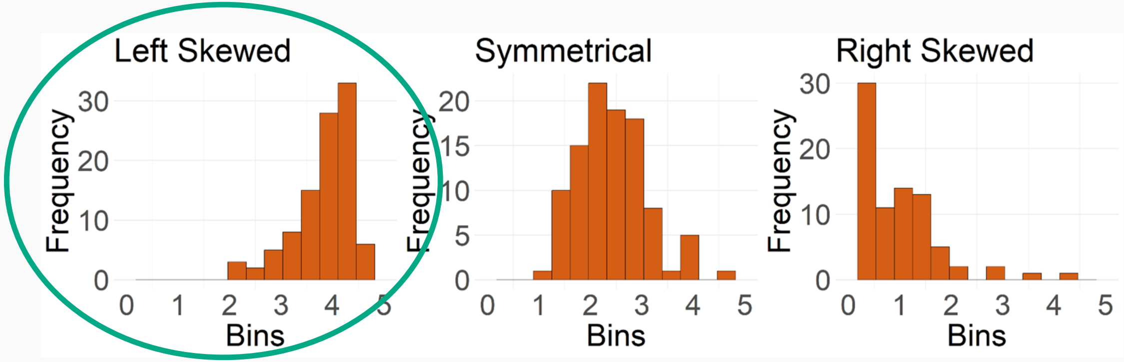


Shape

- Describes a quantitative variable using a number or verbal description
- Summarizes how the data would look in a histogram
- Helps in analyzing data spread and making informed decisions
- Also known as **distribution**



Left Skewed Variables

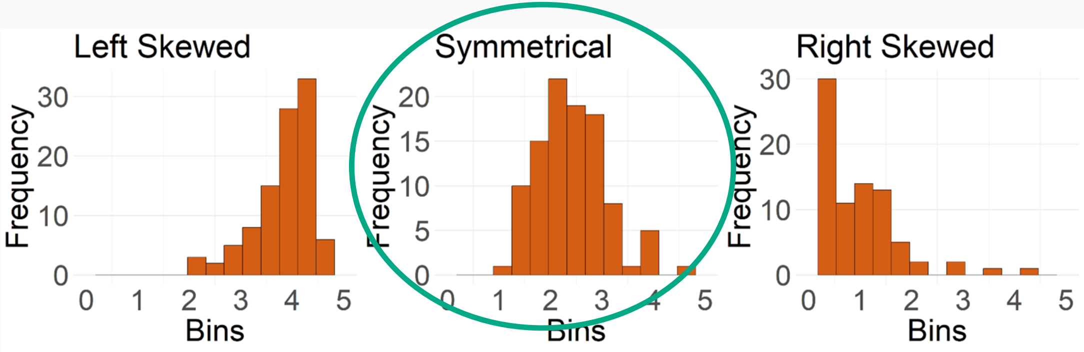


- Most of the observations are medium/large, with a few smaller observations

- Left tail (smaller values) is longer than the right tail (larger values)

Symmetric Variables

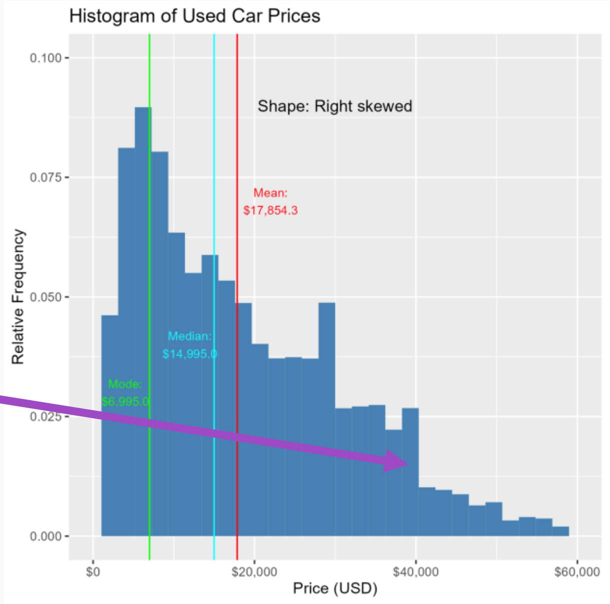
- Observations are evenly distributed on both sides of the center
- Mean and median are near each other
- Shapes on either side of a line drawn down the middle of the graph somewhat mirror one another



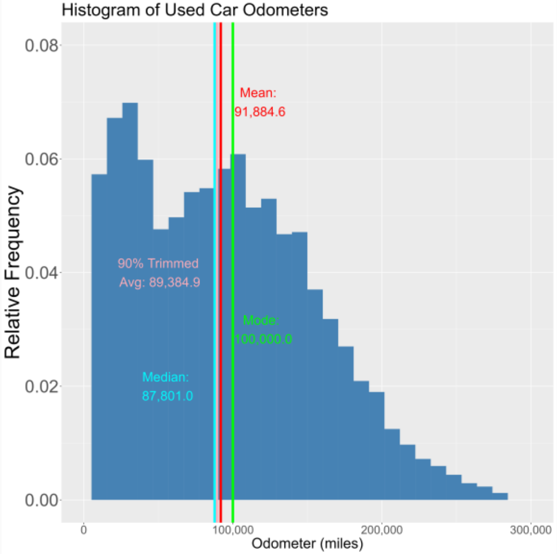
Right Skewed Variables

- Most observations are small, with a few larger observations

- Right tail (larger values) is longer than the left tail (smaller values)



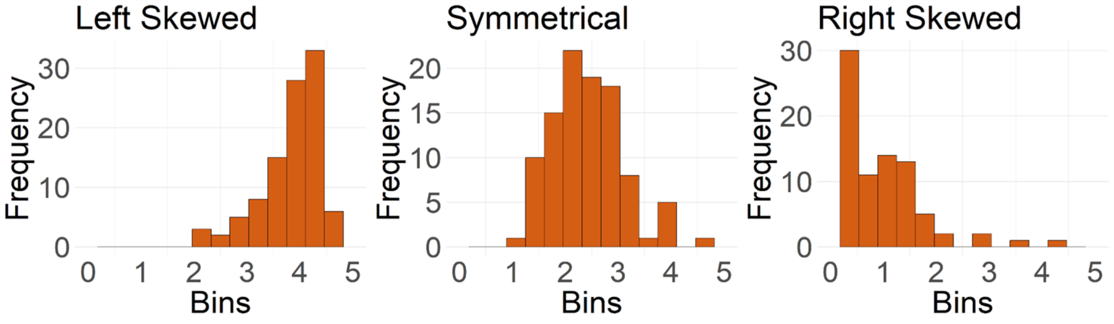
Bimodal Variables



- Histogram has two modes or peaks
- Can be symmetric
- Identifying values of the modes can provide insights

Summary

01 Shape



Shape

Overview

Data Associations

Identifying Outliers

Overview

01 Outliers: The Quartile and Mean/SD Methods

02 Choosing the Best Method for Detecting Outliers

03 Ethics in Practice: Simpson's Paradox

Overview

01 **Outliers: The Quartile and Mean/SD Methods**

Outliers

- Data points that significantly differ from other observations in a dataset
- **Mild outliers:** Moderate differences
- **Regular outliers**: Extreme differences

The Quartile Method

1. Uses first and third quartiles of the data set to detect outliers
2. **The Interquartile Range (IQR):** Q3-Q1, provides a measure of data dispersion
3. **Mild outliers:** Data points below Q1-1.5*IQR or above Q3+1.5*IQR
4. **Regular outliers**: Data points outside Q1-3*IQR or Q3+3*IQR

The Mean/SD Method

1 Uses the mean and standard deviation of a dataset to detect outliers

2 Outliers are values separated from the mean by a certain number of standard deviations

3 **Mild outliers**: 2*SD above or below the mean

4 **Regular outliers**: 3*SD above or below the mean

Overview

02 Choosing the Best Method for Detecting Outliers

Choosing the Best Method for Detecting Outliers

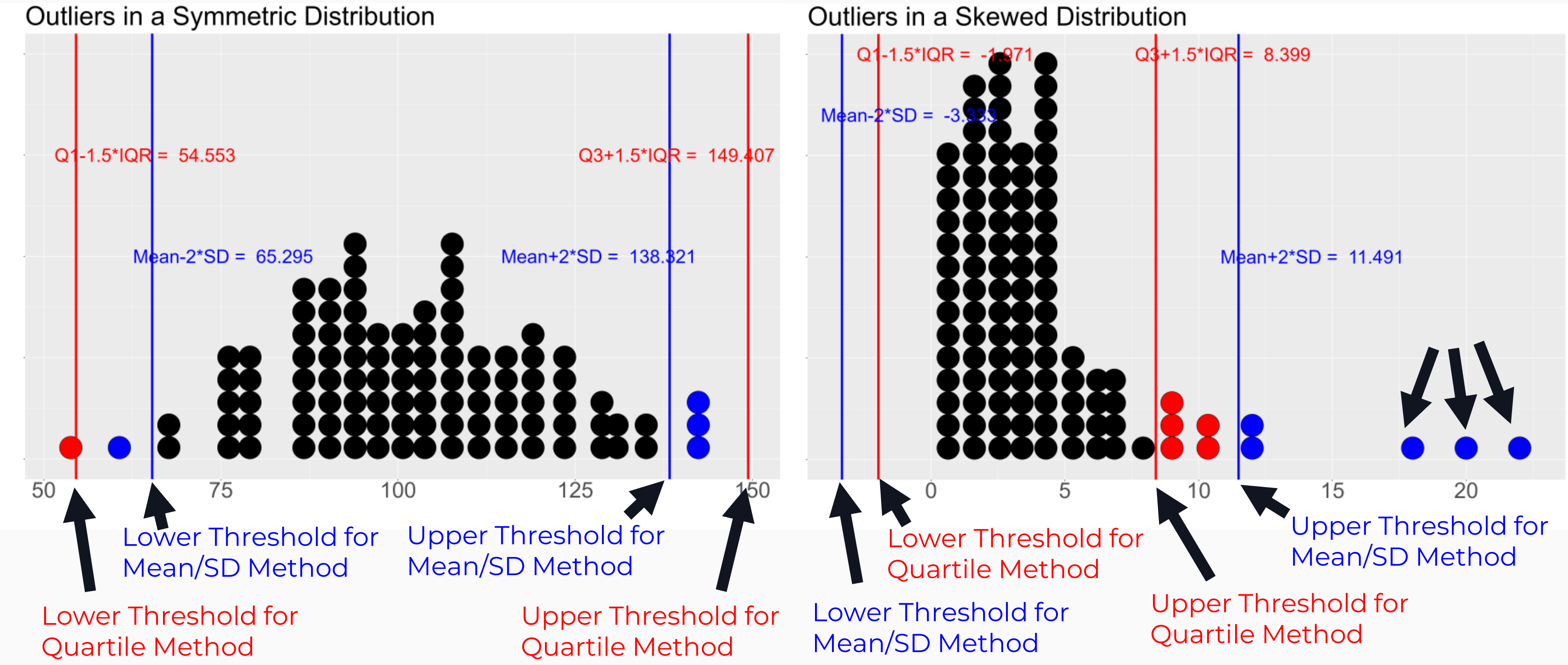
Quartile Method

- Less sensitive to extreme values
- Prefered for skewed data distributions

Mean/SD Method

- Relatively simple
- Better for symmetrical data distributions

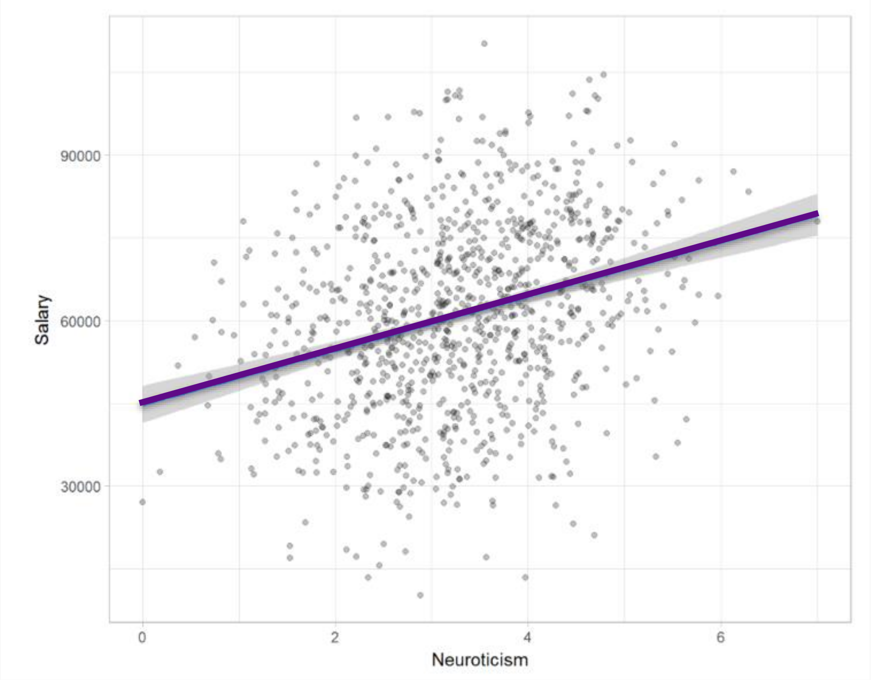
Choosing the Best Method for Detecting Outliers

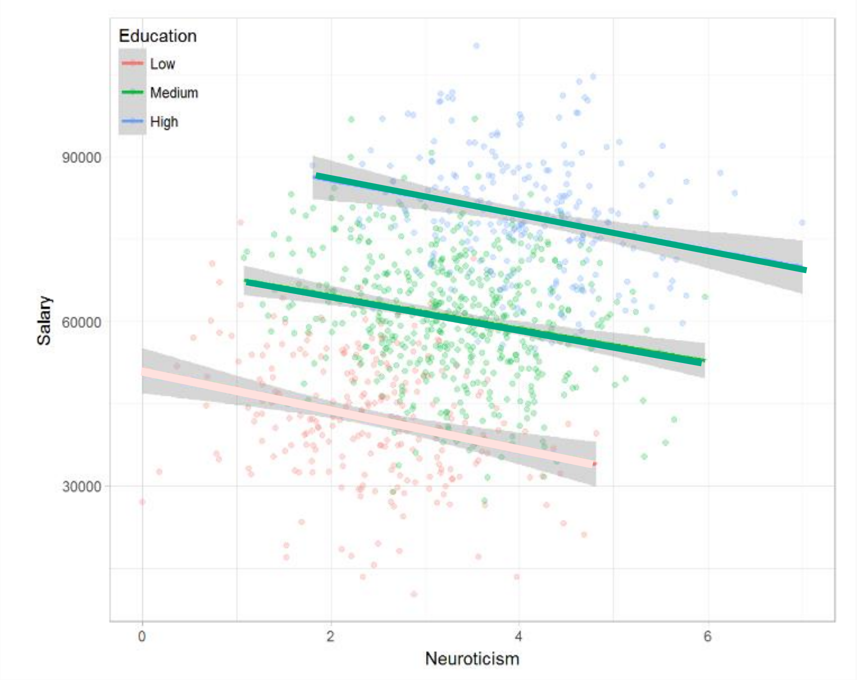


Overview

03 Ethics in Practice: Simpson's Paradox

Ethics in Practice: Simpson's Paradox





Summary

01 Outliers: The Quartile and Mean/SD Methods

02 Choosing the Best Method for Detecting Outliers

03 Ethics in Practice: Simpson’s Paradox

Overview

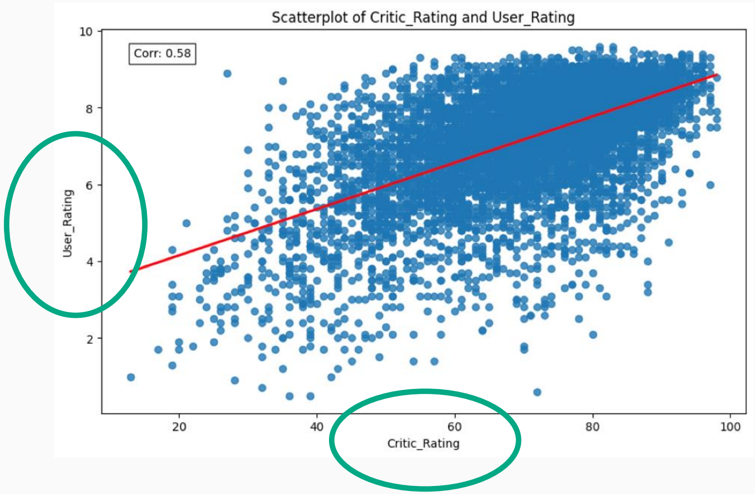
01 **Data Associations**

Data Associations

Relationships between pairs of variables take three primary forms:

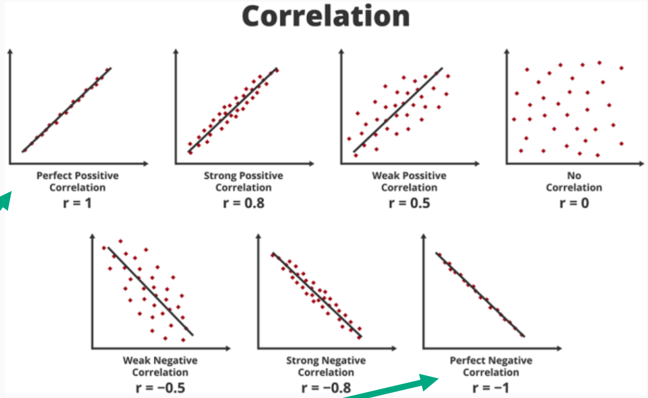
- quantitative-by-quantitative
- quantitative-by-categorical
- categorical-by-categorical

Example: two quantitative variables (quantitative-by-quantitative)



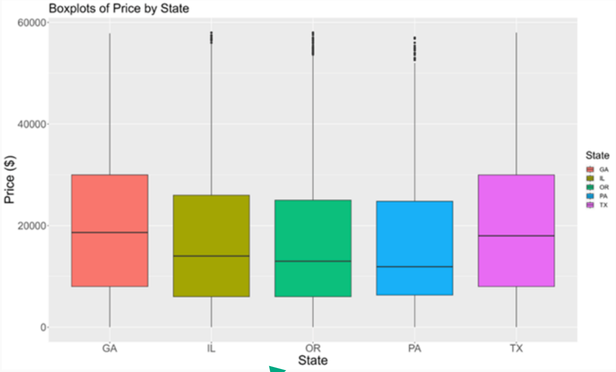
Quantitative-by-Quantitative Relationships

- **Correlation coefficient**: Always between -1 and 1
- Measures **strength** and **direction** of the **linear relationship** between two quantitative variables
- Correlation coefficient of 1: Perfect positive, linear relationship
- Correlation coefficient of -1: Perfect negative, linear relationship



Quantitative-by-Categorical Relationships

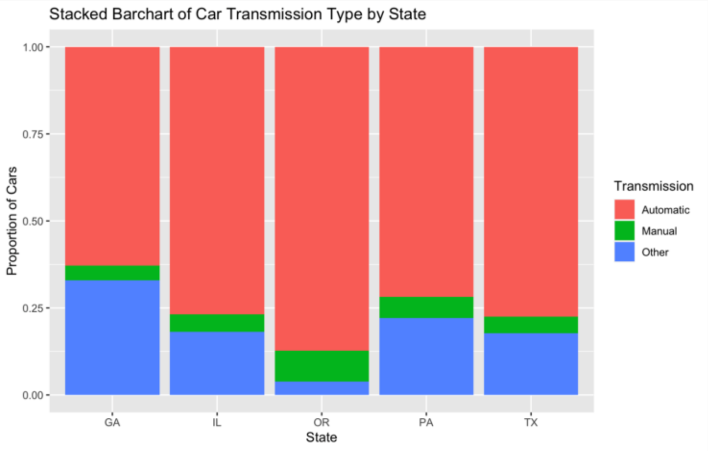
State	Mean Price (USD)	Median Price (USD)	Standard Deviation of Price (USD)
Georgia	19,779	18,639	12,732
Illinois	16,752	13,998	12,734
Oregon	16,724	12,980	14,075
Pennsylvania	15,960	11,900	12,121
Texas	19,757	17,990	13,290



- Quantitative variable: Price of a car
  Categorical variable: State in which the car is sold
- The table and box plot show the quantitative variable's center and variability across categorical variable values

Categorical-by-Categorical Relationships

- First categorical variable: State in which the car is sold
- Second categorical variable: Type of transmission on the sold car
- The stacked bar chart here shows conditional proportions of transmission type by state

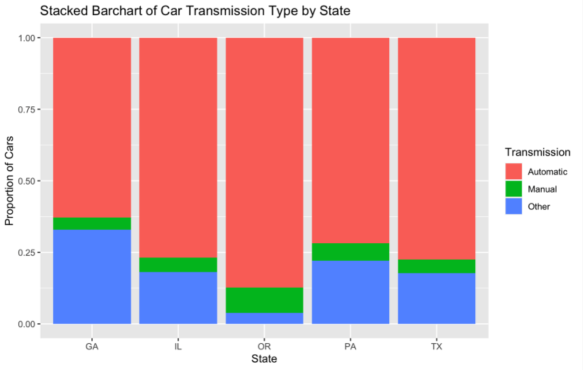


Categorical-by-Categorical Relationships

- **Conditional counts** and **proportions** can be useful summaries when exploring two categorical variables

- The bar chart can be an even better visualization of these two variables!

	Automatic	Manual	Other	Total
GA	3979	285	2016	6280
IL	7551	549	1739	9839
OR	12440	1246	515	14201
PA	9119	837	2770	12726
TX	16554	1117	3643	21314
Total	49643	4034	10683	64363



Big Picture Break: Data Associations

1. Two variables can be *associated* without being *correlated*
    a. An *association* means that there is some type of relationship between two variables
    b. A *correlation* means there is a linear relationship between two quantitative variables described well by a straight line
2. Two variables can be *associated* or *correlated* without having a *causal relationship*
    a. A *causal relationship* between two variables means that changes in one variable will cause changes in the other

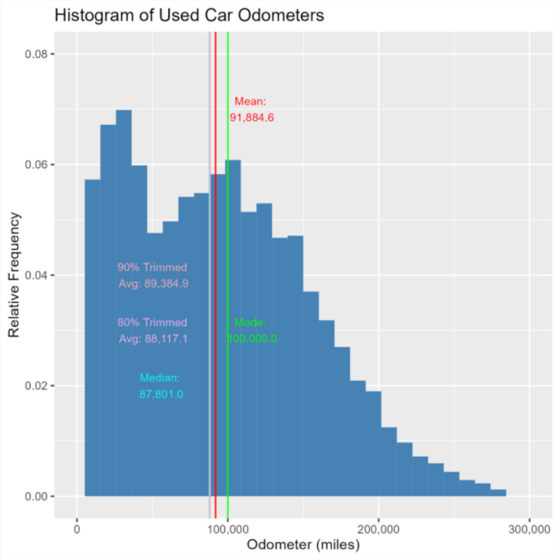
Correlation does not imply causation!

Overview

02 **Percentiles**

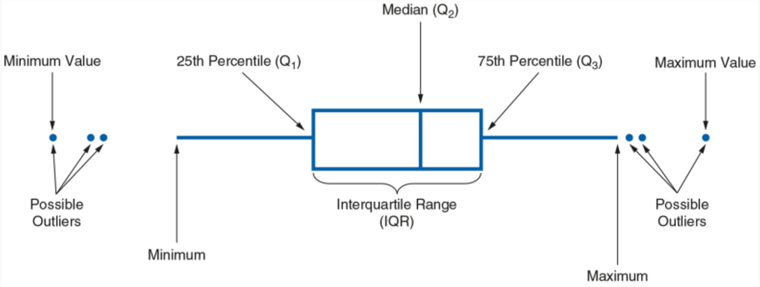
Percentiles

- The *kth* **percentile** for a quantitative variable is the value *at or below which k* percent of the values in its distribution fall
- For example, the 90th percentile of a variable is the value below which 90% of the data points lie



Interquartile Range

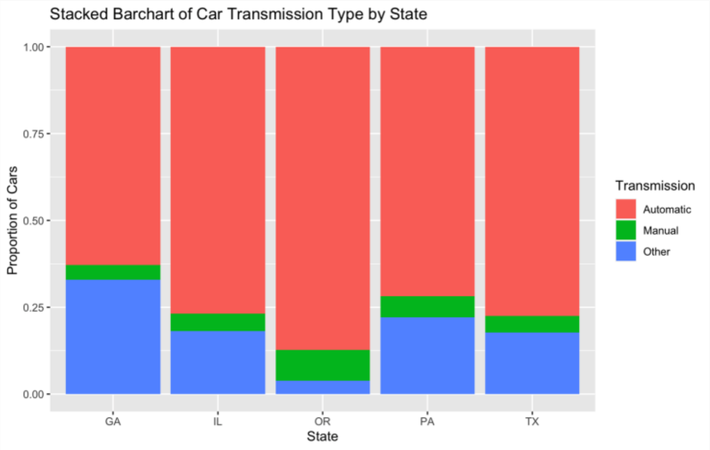
- Difference between a quantitative variable's third and first quartile
- Equivalent to the difference between the 75th percentile and the 25th percentile
- Measures the variability of the middle 50% of a dataset
- More resistant than the range of the entire dataset



Case Study: Exploratory Data Analysis of Electronic Medical Records

- Text mining detects useful patterns and insights in large volumes of unstructured text data
- Psychiatrists analyze results of text-mining medical records to reduce seclusion in psychiatric wards
- Exploratory data analysis supports staff communication, motivates intervention, and enhances patient and staff safety

Summary



Data Associations

Percentiles and IQR